

FINANCIAL ANALYSIS AND PLANNING— RATIO ANALYSIS



LEARNING OUTCOMES

After studying this chapter, you would be able to -

- ♦ Discuss Sources of financial data for Analysis.
- ♦ Discuss financial ratios and its types.
- ♦ Discuss use of financial ratios to analyse the financial statement.
- ♦ Analyse the ratios from the perspective of investors, lenders, suppliers, managers etc. to evaluate the profitability and financial position of an entity.
- ♦ Describe the users and objective of Financial Analysis - A Birds Eye View
- ♦ Discuss Du Pont analysis.
- ♦ State the limitations of Ratio Analysis.

CHAPTER OVERVIEW



RATIO ANALYSIS

Types of Ratios

- ♦ Liquidit Ratios/Short-Solvency
- ♦ Leverag Ratio/Long-Solvency
- ♦ Activit Ratios/Efficienc Ratios/Performance
- Ratios/ Turnover Ratios
- ▲ Profitability Ratios

Application of Ratio Analysis in decision making

Relationship of Management with disciplines of accounting

1. INTRODUCTION

The basis for financial analysis, planning and decision making is financial statements which mainly consist of Balance Sheet and Profit and Loss Account. The profit & loss account shows the operating activities of the concern over a period of time and the balance sheet depicts the balance value of the acquired assets and of liabilities or in other words, financial position of an organization at a particular point of time.

However, the above statements do not disclose all of the necessary and relevant information. For the purpose of obtaining the material and relevant information necessary for ascertaining the financial strengths and weaknesses of an enterprise, it is necessary to analyse the data depicted in the financial statement.

The financial manager has certain analytical tools which help in financial analysis and planning. One of the main tools is Ratio Analysis. Let us discuss the Ratio Analysis in this chapter.



2. RATIO AND RATIO ANALYSIS

Let us first understand the definition of ratio and meaning of ratio analysis.

2.1 Definition of Ratio

A ratio is defined as **“the indicated quotient of two mathematical expressions and as the relationship between two or more things.”** Here, ratio means financial ratio or accounting ratio which is a mathematical expression of the relationship between two accounting figures.

2.2 Ratio Analysis

The term financial ratio can be explained by defining how it is calculated and what the objective of this calculation is?

a. Calculation Basis (Basis of Calculation):

- A relationship expressed in mathematical terms
- Between two individual figures or group of figures
- Connected with each other in some logical manner
- Selected from financial statements of the concern

b. Objective for financial ratios is that all stakeholders (owners, investors, lenders, employees etc.) can draw conclusions about the:

- Performance (past, present and future)
- Strengths & weaknesses of a firm
- Can take decisions in relation to the firm

Ratio analysis is based on the fact that a single accounting figure by itself may not communicate any meaningful information but when expressed relative to some other figure, it may definitely provide some significant information.

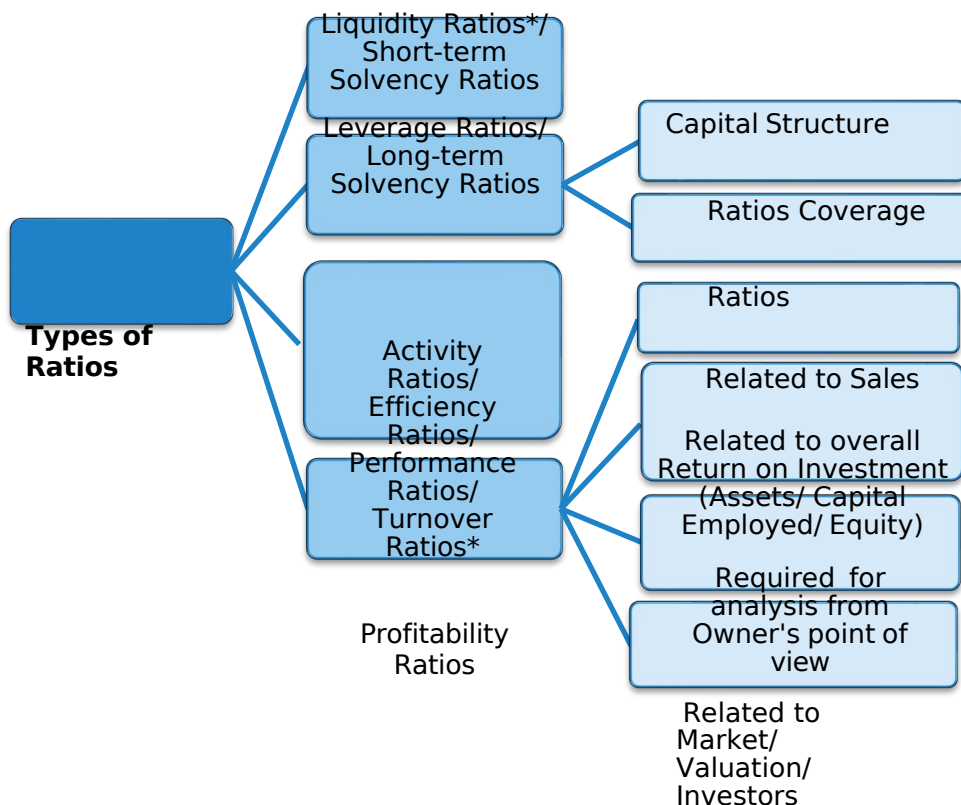
Ratio analysis is not just comparing different numbers from the balance sheet, income statement, and cash flow statement. It is comparing the number against previous years (intra-firm comparison) and, other companies (inter-firm comparison), the industry, or even the economy in general for the purpose of financial analysis.

2.3 Sources of Financial Data for Analysis

The sources of information for financial statement analysis are:

- i. Annual Reports
- ii. Interim financial statements
- iii. Notes to Accounts
- iv. Statement of cash flows
- v. Business periodicals.
- vi. Credit and investment advisory services

3. TYPES OF RATIOS



Classification of Ratios

**Liquidity ratios should be examined taking relevant turnover ratios into consideration.*

3.1 Liquidity Ratios

The terms '**liquidity**' and '**short-term solvency**' are used synonymously.

Liquidity or short-term solvency means ability of the business to pay its short-term liabilities. Inability to pay-off short-term liabilities affects its credibility as well as its credit rating. Continuous default on the part of the business leads to commercial bankruptcy. Eventually such commercial bankruptcy may lead to its sickness and dissolution. Short-term lenders and creditors of a business are very much interested to know its state of liquidity because of their financial stake. Both lack of sufficient liquidity and excess liquidity is bad for the organization.

Various Liquidity Ratios are:

- (a) Current Ratio
- (b) Quick Ratio or Acid test Ratio
- (c) Cash Ratio or Absolute Liquidity Ratio
- (d) Basic Defense Interval or Interval Measure Ratios
- (e) Net Working Capital

(a) Current Ratio: The Current Ratio is one of the best known measures of short-term solvency. It is the most common measure of short-term liquidity.

The main question this ratio addresses is: "**Does your business have enough current assets to meet the payment schedule of its current debts with a margin of safety for possible losses in current assets?**" In other words, current ratio measures whether a firm has enough resources to meet its current obligations.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Where,

Current Asset = Inventories + Sundry Debtors + Cash and Bank Balances + Receivables/ Accruals + Loans and Advances + Disposable Investments + Any other current assets.

Current Liabilities = Creditors for goods and services + Short-term Loans + Bank Overdraft + Cash Credit + Outstanding Expenses
 + Provision for Taxation + Proposed Dividend + Unclaimed Dividend
 + Any other current liabilities.

Interpretation

A generally acceptable current ratio is 2:1. But whether or not a specific ratio is satisfactory depends on the nature of the business and the characteristics of its current assets and liabilities.

- (b) **Quick Ratio:** The Quick Ratio is sometimes called the "**acid-test**" ratio and is one of the best **measures of liquidity**.

$$\text{Quick Ratio or Acid Test Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Where,

$$\begin{aligned} \text{Quick Assets} &= \text{Current Assets} - \text{Inventories} - \text{Prepaid expenses} \\ \text{Current Liabilities} &= \text{As mentioned under Current Ratio.} \end{aligned}$$

The Quick Ratio is a much more conservative measure of short-term liquidity than the Current Ratio. It helps answer the question: "If all sales revenues should disappear, could my business meet its current obligations with the readily convertible quick funds on hand?"

Quick Assets consist of only cash and near cash assets. Inventories are deducted from current assets on the belief that these are not 'near cash assets' and also because in times of financial difficulty, inventory may be saleable only at liquidation value. But in a seller's market, inventories are also near cash assets.

Interpretation

An acid-test of 1:1 is considered satisfactory unless the majority of "quick assets" are in accounts receivable, and the pattern of accounts receivable collection lags behind the schedule for paying current liabilities.

- (c) **Cash Ratio/ Absolute Liquidity Ratio:** The cash ratio measures the absolute liquidity of the business. This ratio considers only the absolute liquidity available with the firm. This ratio is calculated as:

$$\text{Cash Ratio} = \frac{\text{Cash and Bank balances} + \text{Marketable Securities}}{\text{Current Liabilities}}$$

$$= \frac{\text{Cash and Bank balances} + \text{Current Assets}}{\text{Current Liabilities}}$$

Interpretation

The Absolute Liquidity Ratio only tests short-term liquidity in terms of cash and marketable securities/ current investments.

- (d) **Basic Defense Interval/ Interval Measure:**

$$\text{Basic Defense Interval} = \frac{\text{Cash and Bank balances} + \text{Net Marketable Securities}}{\text{Operating Expenses} \div \text{No. of days (say 360)}}$$

$$= \frac{\text{Current Assets} - \text{Prepaid expenses} - \text{Accounts Payable}}{\text{Daily Operating Expenses}}$$

$$\text{Daily Operating Expenses} = \frac{\text{Cost of Goods Sold} + \text{Selling and Administrative Expenses} + \text{non cash expenditure}}{\text{No. of days in a year}}$$

Interpretation

If for some reason all the company's revenues were to suddenly cease, the Basic Defense Interval would help determine the number of days for which the company can cover its cash expenses without the aid of additional financing.

- (e) **Net Working Capital:** Net working capital is more a measure of cash flow than a ratio. The result of this calculation must be a positive number. However, in certain business models it may be negative. It is calculated as shown below:

$$\text{Net Working Capital (Excluding)} = \text{Current Assets} - \text{Current Liabilities}$$

Interpretation

Bankers look at Net Working Capital over time to determine a company's ability to weather financial crises. Loans are often tied to minimum working capital requirements.

3.2 Long-term Solvency Ratios/ Leverage Ratios

The leverage ratios may be defined as those financial ratios which measure the **long-term stability and capital structure of the firm**. These ratios indicate the mix of funds provided by owners and lenders and assure the lenders of the long-term funds with regard to:

- (i) Periodic payment of interest during the period of the loan and
- (ii) Repayment of principal amount on maturity.

Leverage ratios are of two types:

1. Capital Structure Ratios

- (a) Equity Ratio
- (b) Debt Ratio
- (c) Debt to Equity Ratio
- (d) Debt to Total Assets Ratio
- (e) Capital Gearing Ratio
- (f) Proprietary Ratio

2. Coverage Ratios

- (a) Debt-Service Coverage Ratio (DSCR)
- (b) Interest Coverage Ratio
- (c) Preference Dividend Coverage Ratio
- (d) Fixed Charges Coverage Ratio

3.2.1 Capital Structure Ratios

These ratios provide an insight into the financing techniques used by a business and focus, as a consequence, on the **long-term solvency position**.

From the balance sheet, one can get only the absolute fund employed and its sources but only capital structure ratios show the relative weight of different sources.

Various capital structure ratios are:

(a) Equity Ratio:

$$\text{Equity Ratio} = \frac{\text{Shareholder's}}{\text{Net}}$$

The **shareholder's equity** is Equity share capital and Reserves & Surplus (excluding fictitious assets etc).

Net Assets or Capital employed includes Net Fixed Assets and Net Current Assets (Current Assets – Current Liabilities).

This ratio indicates proportion of owner's fund to total fund invested in the business. Traditionally, it is believed that higher the proportion of owner's fund, lower is the degree of risk for potential lenders.

(b) Debt Ratio:

$$\text{Debt Ratio} = \frac{\text{Total}}{\text{Net}}$$

Total debt or total outside liabilities includes short and long term borrowings from financial institutions, debentures/bonds, deferred payment arrangements for buying capital equipment, bank borrowings, public deposits and any other interest bearing loan.

Interpretation

This ratio is used to analyse the long-term solvency of a firm. A ratio greater than 1 would mean greater portion of company assets are funded by debt and could be a risky scenario.

(c) Debt to Equity Ratio:

$$\begin{aligned} \text{Debt to Equity Ratio} &= \frac{\text{Total Outside Liabilities}}{\text{Shareholders' Equity}} = \frac{\text{Total Debt}^*}{\text{Shareholder's Equity}} \\ &= \frac{\text{Long-term Debt}^*}{\text{Shareholders' equity}} \end{aligned}$$

*Not merely long-term debt i.e. both current & non-current liabilities.

** Sometimes only interest-bearing, long-term debt is used instead of total liabilities (exclusive of current liabilities)

Interpretation

A high debt to equity ratio here means less protection for creditors, a low ratio, on the other hand, indicates a wider safety cushion (i.e., creditors feel the owner's funds can help absorb possible losses of income and capital). This ratio indicates the proportion of debt fund in relation to equity. This ratio is very often used for making capital structure decisions such as issue of shares and/ or debentures. Lenders are also very keen to know this ratio since it shows relative weights of debt and equity. Debt equity ratio is the indicator of firm's financial leverage.

(d) Debt to Total Assets Ratio: This ratio measures the **proportion of total assets financed with debt** and, therefore, the extent

$$\text{Debt to Total Assets Ratio} = \frac{\text{Total Outside Liabilities}}{\text{Total}} \quad \text{Or} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

of financial leverage.

Higher the ratio, indicates that assets are less backed up by equity and hence higher financial leverage.

- (e) **Capital Gearing Ratio:** In addition to debt-equity ratio, sometimes capital gearing ratio is also calculated to show the proportion of fixed interest (dividend) bearing capital to funds belonging to equity shareholders i.e. equity funds or net worth. Again, higher ratio

$$\text{Capital Gearing Ratio} = \frac{\text{Preference Share Capital} + \text{Debentures} + \text{Other}}{\text{Equity Share Capital} + \text{Reserves \& Surplus}}$$

may indicate more risk.

- (f) **Proprietary Ratio:**

$$\text{Proprietary Ratio} = \frac{\text{Proprietary}}{\text{Total}}$$

Proprietary fund includes Equity Share Capital, Preference Share Capital and Reserve & Surplus.

Total assets exclude fictitious assets and losses.

Interpretation

It indicates the proportion of total assets financed by shareholders. Higher the ratio, less risky scenario it shall be.

3.2.2 Coverage Ratios

The coverage ratios measure the **firm's ability to service the fixed liabilities**. These ratios establish the relationship between fixed claims and what is normally available out of which these claims are to be paid. The fixed claims consist of:

- (i) Interest on loans
- (ii) Preference dividend
- (iii) Amortisation of principal or repayment of the instalment of loans or redemption of preference capital on maturity.

The following are important coverage ratios:

- (a) **Debt Service Coverage Ratio (DSCR):** Lenders are interested in **debt service coverage to judge the firm's ability to pay off current interest and instalments.**

$$\text{Debt Service Coverage Ratio} = \frac{\text{Earnings available for debt}}{\text{Interest} +}$$

Earnings available for debt service*

= Net profit (Earning after taxes) + Non-cash operating expenses like depreciation and other amortizations + Interest + other adjustments like loss on sale of Fixed Asset etc.

*Fund from operations (or cash from operations) before interest and taxes also can be considered as per the requirement.

Interpretation

Normally DSCR of 1.5 to 2 is satisfactory. You may note that sometimes in both numerator and denominator lease rentals may also be added.

- (b) **Interest Coverage Ratio:** This ratio also known as “**times interest earned ratio**” indicates the firm’s ability to meet interest (and other fixed charges) obligations. This ratio is computed as:

$$\text{Interest Coverage Ratio} = \frac{\text{Earnings before interest and taxes (EBIT)}}{\text{Interest}}$$

Interpretation

Earnings before interest and taxes are used in the numerator of this ratio because the ability to pay interest is not affected by tax burden as interest on debt funds is deductible expense. It measures how many times a company can cover its current interest payment with its available earnings? In other words, it reflects the margin of safety a company has for paying interest on its debt during a given period.

A high interest coverage ratio means that an enterprise can easily meet its interest obligations even if earnings before interest and taxes suffer a considerable decline. A lower ratio indicates excessive use of debt or inefficient operations.

- (c) **Preference Dividend Coverage Ratio:** This ratio measures the **ability of a firm to pay dividend on preference shares** which carry a stated rate of return. This ratio is computed as:

$$\text{Preference Dividend Coverage Ratio} = \frac{\text{Net Profit/Earning after}}{\text{Preference}}$$

Interpretation

This ratio indicates margin of safety available to the preference shareholders. A higher ratio is desirable from preference shareholders point of view.

Similarly, **Equity Dividend coverage ratio** can also be calculated as:

$$\text{Equity Dividend Coverage Ratio} = \frac{\text{Earning after taxes (EAT)} - \text{Equity}}{\text{Equity}}$$

- (d) **Fixed Charges Coverage Ratio:** This ratio shows how many times the cash flow before interest and taxes covers all fixed financing charges. This ratio of more than 1 is considered as safe.

$$\text{Fixed Charges Coverage Ratio} = \frac{\text{EBIT} - \text{Depreciation}}{\text{Interest} + \text{Repayment of}}$$

Notes for calculating Ratios:

1. EBIT (Earnings before interest and taxes) = PBIT (Profit before interest and taxes),
EAT (Earnings after taxes) = PAT (Profit after taxes)
EBT (Earnings before taxes) = PBT (Profit before taxes)
2. Ratios shall be calculated based on requirement and availability of information and may deviate from original formulae. If required, assumptions should be given.
3. Numerator should be taken in correspondence with the denominator and vice-versa.

3.3 Activity Ratios/ Efficiency Ratios/ Performance Ratios/ Turnover Ratios

These ratios are employed to **evaluate the efficiency with which the firm manages and utilises its assets**. For this reason, they are often called as 'Asset management ratios'. These ratios usually indicate the frequency of sales with

respect to its assets. These assets may be capital assets or working capital or average inventory.

Activity Ratios/ Efficiency Ratios/ Performance Ratios/ Turnover Ratios:

- (a) Total Assets Turnover Ratio
- (b) Fixed Assets Turnover Ratio
- (c) Capital Turnover Ratio/ Net Assets Turnover Ratio
- (d) Current Assets Turnover Ratio
- (e) Working Capital Turnover Ratio
 - (i) Inventory/ Stock Turnover Ratio
 - (ii) Receivables (Debtors) Turnover Ratio
 - (iii) Payables (Creditors) Turnover Ratio

These ratios are usually calculated with reference to **sales/cost of goods sold** and are expressed in terms of rate or times.

- (a) Total Asset Turnover Ratio:** This ratio measures the efficiency with which the firm uses its total assets. Higher the ratio, better it is. This ratio is computed as:

$$\text{Total Asset Turnover Ratio} = \frac{\text{Sales/ Cost of}}{\text{Total Asset}}$$

Interpretation

A high total assets turnover ratio indicates the efficient utilisation of total assets in generation of sales. Similarly, a low asset turnover ratio indicates total assets are not efficiently used to generate sales.

- (b) Fixed Assets Turnover Ratio:** It measures the efficiency with which the firm uses its fixed assets.

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Sales/ Cost of}}{\text{Fixed Asset}}$$

Interpretation

A high fixed assets turnover ratio indicates efficient utilisation of fixed assets in generating sales. A firm whose plant and machinery are old may show a higher fixed assets turnover ratio than the firm which has purchased them recently.

(c) **Capital Turnover Ratio/ Net Asset Turnover Ratio:**

$$\text{Capital Turnover Ratio} = \frac{\text{Sales/ Cost of}}{\text{Net}}$$

Interpretation

Since Net Assets equals to capital employed it is also known as Capital Turnover Ratio. This ratio indicates the firm's ability of generating sales/ Cost of Goods Sold per rupee of long-term investment. The higher the ratio, the more efficient is the utilisation of owner's and long-term creditors' funds.

(d) **Current Assets Turnover Ratio:** It measures the efficiency of using the current assets by the firm.

$$\text{Current Assets Turnover Ratio} = \frac{\text{Sales/ Cost of}}{\text{Current}}$$

Interpretation

The higher the ratio, the more efficient is the utilisation of current assets in generating sales.

(e) **Working Capital Turnover Ratio:** It measures how effective a company is at generating sales for every rupee of working capital

$$\text{Working Capital Turnover Ratio} = \frac{\text{Sales/ Cost of}}{\text{Working}}$$

put to use.

Interpretation

Higher the ratio, the more efficient is the utilisation of working capital in generating sales. However, a very high working capital turnover ratio indicates that the company needs to raise additional working capital for future needs.

Working Capital Turnover is further segregated into Inventory Turnover, Debtors Turnover, and Creditors Turnover.

Note: Average of Total Assets/ Fixed Assets/ Current Assets/ Net Assets/ Working Capital also can be taken in the denominator for the above ratios.

- (i) **Inventory/ Stock Turnover Ratio:** This ratio also known as **stock turnover ratio establishes the relationship between the cost of goods sold during the year** and average inventory held during the year. It measures the efficiency with which a firm utilizes or manages its inventory. It is calculated as follows:

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average}}$$

$$\text{Average Inventory} = \frac{\text{Opening Stock} + \text{Closing Stock}}{2}$$

In the case of inventory of raw material, the inventory turnover ratio is calculated using the following formula :

$$\text{Raw Material Inventory Turnover Ratio} = \frac{\text{Raw Material Consumed}}{\text{Average Raw Material}}$$

Interpretation

This ratio indicates that how fast inventory is used or sold. A high ratio is good from the view point of liquidity and vice versa. A low ratio would indicate that inventory is not used/ sold/ lost and stays in a shelf or in the warehouse for a long time.

- (ii) **Receivables (Debtors) Turnover Ratio:** In case firm sells goods on credit, the realization of sales revenue is delayed and the receivables are created. The cash is realised from these receivables later on.

The **speed with which these receivables are collected affects** the liquidity position of the firm. The debtor's turnover ratio throws light on the collection and credit policies of the firm. **It measures the efficiency with which management is managing its accounts receivables.** It is calculated as follows:

$$\text{Receivables (Debtors) Turnover Ratio} = \frac{\text{Credit Sales}}{\text{Average Accounts}}$$

A low debtors' turnover ratio reflects liberal credit terms granted to customers, while a high ratio shows that collections are made rapidly.

Receivables (Debtors) Velocity/Average Collection Period:

Debtor's turnover ratio indicates the average collection period. However, the average collection period can be directly calculated as

$$= \frac{\text{Average Accounts}}{\text{Average Daily Credit}} \quad \text{Or} \quad \frac{12 \text{ months}/52 \text{ weeks}/360}{\text{days Receivable}}$$

follows:

$$\text{Average Daily Credit Sales} = \frac{\text{Credit Sales}}{\text{No. of days in year (say 360)}}$$

Interpretation

The average collection period measures the average number of days it takes to collect an account receivable. This ratio is also referred to as the number of days of receivable and the number of day's sales in receivables. In determining the credit policy, debtor's turnover and average collection period provide a unique guidance.

- (iii) Payables Turnover Ratio:** This ratio is calculated on the same lines as receivable turnover ratio is calculated. It measures how fast a company makes payment to its creditors. It shows the velocity of payables payment by the firm. It is calculated as follows:

$$\text{Payables Turnover Ratio} = \frac{\text{Annual Net Credit}}{\text{Average Accounts}}$$

A low creditor's turnover ratio reflects liberal credit terms granted by suppliers, while a high ratio shows that accounts are settled rapidly.

Payable Velocity/ Average payment period can be calculated using:

$$= \frac{\text{Average Accounts Payable}}{\text{Average Daily Credit}} \quad \text{Or} \quad \frac{12 \text{ months}/52 \text{ weeks}/}{\text{Payables}}$$

Interpretation

The firm can compare what credit period it receives from the suppliers and what it offers to the customers. Also, it can compare the average credit period offered to the customers in the industry to which it belongs.

The above three ratios i.e. Inventory Turnover Ratio/ Receivables Turnover Ratio/Payables Turnover Ratio are also relevant to examine liquidity of an organization.

Notes for calculating Ratios:

1. Only selling & distribution expenses differentiate Cost of Goods Sold (COGS) and Cost of Sales (COS). In its absence, COGS will be equal to Cost of Sales.
2. We can consider Cost of Goods Sold/ Cost of Sales to calculate turnover ratios eliminating profit part.
3. Average of Total Assets/ Fixed Assets/ Current Assets/ Net Assets/ Working Capital also can be taken in denominator while calculating the above ratios. In fact, when average figures of total assets, net assets, capital employed, shareholders' fund etc. are available it may be preferred to calculate ratios by using this information.
4. Ratios shall be calculated based on requirement and availability of information and may deviate from original formulae. If required, assumptions should be given.

3.4 Profitability Ratios

The profitability ratios **measure the profitability or the operational efficiency** of the firm. These ratios reflect the final results of business operations. They are some of the most closely watched and widely quoted ratios. Management attempts to maximize these ratios to maximize the firm's value.

The results of the firm can be evaluated in terms of its earnings with reference to a given level of assets or sales or owner's interest etc. Therefore, the profitability ratios are broadly classified in four categories:

- (i) Profitability Ratios related to Sales
- (ii) Profitability Ratios related to overall Return on Investment

- (iii) Profitability Ratios required for Analysis from Owner's Point of View
- (iv) Profitability Ratios related to Market/ Valuation/ Investors

Profitability Ratios are as follows:

(i) Profitability Ratios based on Sales

- (a) Gross Profit Ratio
- (b) Net Profit Ratio
- (c) Operating Profit Ratio
- (d) Expenses Ratio

(ii) Profitability Ratios related to Overall Return on Assets/ Investments

- (a) Return on Investments (ROI)
 - (i) Return on Assets (ROA)
 - (ii) Return of Capital Employed (ROCE)
 - (iii) Return on Equity (ROE)

(iii) Profitability Ratios required for Analysis from Owner's Point of View

- (a) Earnings per Share (EPS)
- (b) Dividend per Share (DPS)
- (c) Dividend Pay-out Ratio (DP)

(iv) Profitability Ratios related to Market/ Valuation/ Investors

- (a) Price Earnings (P/E) Ratio
- (b) Dividend and Earning Yield
- (c) Market Value/ Book Value per Share (MV/BV)
- (d) Q Ratio

3.4.1 Profitability Ratios based on Sales

- (a) **Gross Profit (G.P) Ratio/ Gross Profit Margin:** It measures the percentage of each sale in rupees remaining after payment for

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Sale}}$$

the goods sold.

Interpretation

Gross profit margin depends on the relationship between sales price, volume and costs. A high Gross Profit Margin is a favourable sign of good management.

- (b) **Net Profit Ratio/ Net Profit Margin:** It measures the relationship between net profit and sales of the business. Depending on the concept of net profit, it can be calculated as:

$$\begin{aligned} \text{(i) Net Profit Ratio} &= \frac{\text{Net Profit}}{\text{Sale}} \\ &\text{Or} \\ &\frac{\text{Earnings after taxes}}{\text{(EAT) - Sale}} \end{aligned}$$

$$\text{(ii) Pre-tax Profit Ratio} = \frac{\text{Earnings before taxes}}{\text{Sale}}$$

Interpretation

Net Profit ratio finds the proportion of revenue that finds its way into profits after meeting all expenses. A high net profit ratio indicates positive returns from the business.

- (c) **Operating Profit Ratio:**

Operating profit ratio is also calculated to evaluate operating performance of business.

$$\text{Operating Profit Ratio} = \frac{\text{Operating Profit}}{\text{Sale}}$$

or,

$$\frac{\text{Earnings before interest and taxes (EBIT)}}{\text{Sale}}$$

Where,

Operating Profit = Sales - Cost of Goods Sold (COGS) - Operating Expenses

Interpretation

Operating profit ratio measures **the percentage of each sale in rupees that remains after the payment of all costs and expenses except for interest and taxes**. This ratio is followed closely by analysts because it focuses on operating results. Operating profit is often referred to as earnings before interest and taxes or EBIT.

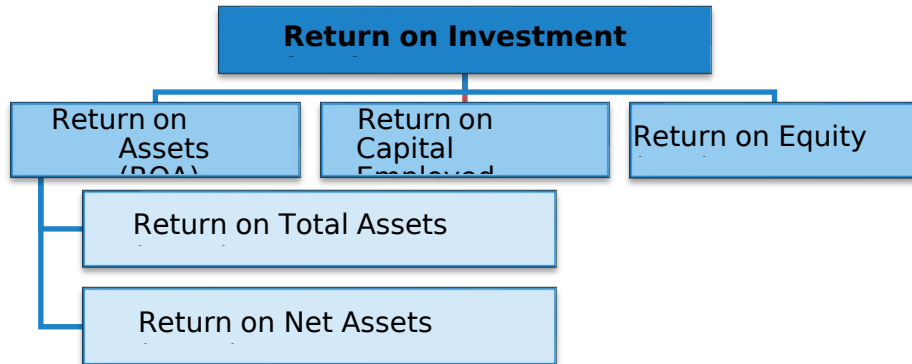
(d) Expenses Ratio: Based on different concepts of expenses it can be expressed in different variants as below:

(i) Cost of Goods Sold (COGS) Ratio	$= \frac{\text{COGS}}{\text{Sales}}$
(ii) Operating Expenses	$= \frac{\text{Administrative} + \text{Selling \& Distribution}}{\text{Sale}}$
(iii) Operating	$= \frac{\text{COGS} + \text{Operating}}{\text{Sale}}$
(iv) Financial Expenses	$= \frac{\text{Financial}}{\text{Sale}}$

*It **excludes** taxes, loss due to theft, goods destroyed by fire etc.

Administration Expenses Ratio and Selling & Distribution Expenses Ratio can also be calculated in similar ways.

3.4.2 Profitability Ratios related to Overall Return on Assets/ Investments



- (a) **Return on Investment (ROI):** ROI is the most important ratio of all. It is the **percentage of return on funds invested in the business by its owners**. In short, this ratio tells the owner whether or not all the effort put into the business has been worthwhile. It compares earnings/ returns/ profit with the investment in the company. The ROI is calculated as follows:

$$\text{Return on Investment} = \frac{\text{Return/Profit/Earnings}}{\text{Investment}} \times 100$$

Or,

$$= \frac{\text{Return/Profit/Earnings}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Investment}}$$

Or,

$$= \text{Profitability Ratio} \times \text{Investment Turnover Ratio}$$

Since, Profitability Ratio = $\frac{\text{Return/Profit/Earnings}}{\text{Sales}}$, and

$$\text{Investment Turnover Ratio} = \frac{\text{Sales}}{\text{Investments}}$$

ROI can be improved either by improving Profitability Ratio or Investment Turnover Ratio or by both.

The concept of investment varies and accordingly there are three broad categories of ROI i.e.

- (i) **Return on Assets (ROA),**
- (ii) **Return on Capital Employed (ROCE) and**
- (iii) **Return on Equity (ROE).**

We should keep in mind that investment may be Total Assets or Net Assets. Further, funds employed in net assets are also known as capital employed which is nothing but Net worth plus Debt, where Net worth is equity shareholders' fund. Similarly, the concept of returns/ earnings/ profits may vary as per the requirement and availability of information.

- (i) **Return on Assets (ROA):** The profitability ratio is measured in terms of relationship between **net profits and assets employed** to earn that profit. This ratio measures the profitability of the firm in terms of assets employed in the firm. Based on various concepts of net profit (return) and assets, the ROA may be measured as follows:

$$\text{ROA} = \frac{\text{Net Profit after taxes}}{\text{* Average Total Assets}} \quad \text{or} \quad \frac{\text{Net Profit after taxes}}{\text{Average Tangible Assets}}$$

$$\quad \quad \quad \text{or} \quad \frac{\text{Net Profit after taxes}}{\text{Average Fixed Assets}}$$

* Note: Sometimes, total assets may also be considered instead of average assets.

Here, net profit is exclusive of interest. As Assets are also financed by lenders, hence ROA can be calculated as:

$$\text{RoA} = \frac{\text{Net Profit after taxes} + \text{Interest}}{\text{Average Total Assets/Average Tangible Assets/Average Fixed Assets}}$$

$$= \frac{\text{EBIT}(1-t)}{\text{Average Total}} \quad \text{\{also known as **Return on Total Assets**\}}$$

Or

$$= \frac{\text{EBIT}(1-t)}{\text{Average Net}} \quad \text{\{also known as **Return on Net Assets**\}}$$

(ii) Return on Capital Employed (ROCE): It is another variation of ROI. The ROCE is calculated as follows:

$$\text{ROCE (Pre-tax)} = \frac{\text{Earnings before interest and tax}}{\text{Capital}}$$

$$\text{ROCE (Post-tax)} = \frac{\text{EBIT}(1-t)}{\text{Capital}}$$

Sometimes, it is also calculated as:

$$= \frac{\text{Net Profit after taxes}}{\text{Capital}}$$

Where,

$$\text{Capital Employed} = \text{Total Assets} - \text{Current Liabilities}$$

Or

$$= \text{Fixed Assets} + \text{Working Capital}$$

Or

$$= \text{Equity} + \text{Long Term Debt}$$

Interpretation

ROCE should always be higher than the rate at which the company borrows.

Intangible assets (assets which have no physical existence like goodwill, patents and trade-marks) should be included in the capital employed. But no fictitious asset (such as deferred expenses) should be included within capital employed. If information is available, then average capital employed shall be taken.

(iii) Return on Equity (ROE): Return on Equity measures the **profitability of equity funds invested in the firm**. This ratio reveals how profitably of the owners' funds have been utilised by the firm. It also measures the percentage return generated to equity shareholders. This ratio is computed as:

$$ROE = \frac{\text{Net Profit after taxes - Preference dividend (if)}}{\text{Net Worth/ Equity}}$$

Interpretation

Return on equity is one of the most important indicators of a firm's profitability and potential growth. Companies that boast a high return on equity with little or no debt are able to grow without large capital expenditures, allowing the owners of the business to withdraw cash and reinvest it elsewhere. Many investors fail to realize, however, that two companies can have the same return on equity, yet one can be a much better business. If return on total shareholders (i.e. equity and preference shareholder) is calculated, then Net Profit after taxes (before preference dividend) shall be divided by total shareholders' fund including preference share capital.

Return on Equity using the Du Pont Model:

A finance executive at E.I. Du Pont de Nemours and Co., of Wilmington, Delaware, created the DuPont system of financial analysis in 1919. That system is used around the world today and serves as the basis of components that make up return on equity.

There are various components in the calculation of return on equity using the traditional DuPont model- the net profit margin, asset turnover, and the equity multiplier. By examining each input individually, the sources of a company's return on equity can be discovered and compared to its competitors. The components are as follows:

- (i) **Profitability/Net Profit Margin:** The **net profit margin is simply the after-tax profit a company generates for each rupee of revenue.** Net profit margin varies across industries, making it important to compare a potential investment against its competitors. Although the general rule-of-thumb is that a higher net profit margin is preferable, it is not uncommon for management to purposely lower the net profit margin in a bid to attract higher sales.

$$\text{Profitability/ Net Profit margin} = \frac{\text{Profit/ Net Income}}{\text{Sales/ Revenue}}$$

Net profit margin is a safety cushion; the lower the margin, the less room for an error. A business with 1% margin has no room for flawed execution. Small miscalculations on management's part could lead to tremendous losses with little or no warning.

- (ii) **Investment Turnover/ Asset Turnover/ Capital Turnover:** The asset turnover ratio is a measure of **how effectively a company converts its assets into sales**. It is calculated as follows:

$$\text{Investment Turnover/ Asset Turnover/ Capital Turnover} = \frac{\text{Sales/ Revenue}}{\text{Investment/ Assets/ Capital}}$$

The asset turnover ratio tends to be inversely related to the net profit margin i.e. higher the net profit margin, lower the asset turnover and vice versa. The result is that the investor can compare companies using different models (low-profit, high-volume vs. high-profit, low-volume) and determine which one is the more attractive business.

- (iii) **Equity Multiplier:** It is possible for a company with terrible sales and margins to take on excessive debt and artificially increase its return on equity. The equity multiplier, a measure of financial leverage, allows the investor to see what portion of the return on equity is the result of debt. The equity multiplier is calculated as

$$\text{Equity Multiplier} = \frac{\text{Investment /Assets}}{\text{Shareholders'}}$$

follows:

Calculation of Return on Equity

To calculate the return on equity using the DuPont model, simply multiply the three components (net profit margin, asset turnover, and

$$\text{Return on Equity} = (\text{Profitability/ Net profit margin}) \times (\text{Investment Turnover/ Asset Turnover / Capital Turnover}) \times \text{Equity multiplier.}$$

Example - 1: XYZ Company's details are as under:

Revenue: ` 29,261; Net Income: ` 4,212; Assets: ` 27,987; Shareholders' Equity: ` 13,572.

Here, Return on Equity as per Du Pont Model will be calculated as follows: Net Profit Margin = Net Income (₹ 4,212) ÷ Revenue (₹ 29,261)

$$= 0.14439 \text{ or } 14.39\%$$

$$\text{Asset Turnover} = \text{Revenue (₹ 29,261)} \div \text{Assets (₹ 27,987)} = 1.0455$$

$$\text{Equity Multiplier} = \text{Assets (₹ 27,987)} \div \text{Shareholders' Equity (₹ 13,572)} = 2.0621$$

Finally, we multiply the three components together to calculate the return on equity:

$$\begin{aligned} \text{Return on Equity} &= \text{Net Profit Margin} \times \text{Asset Turnover} \times \text{Equity Multiplier} \\ &= (0.1439) \times (1.0455) \times (2.0621) = 0.3102, \text{ or } 31.02\% \end{aligned}$$

Analysis: A 31.02% return on equity is good in any industry. Yet, if you were to leave out the equity multiplier to see how much company would earn if it were completely debt-free, you will see that the ROE drops to 15.04% (0.1439 x 1.0455). 15.04% of the return on equity was due to profit margins and sales, while remaining 15.98% was due to returns earned on the debt at work in the business. If you could find a company at a comparable valuation with the same return on equity yet a higher percentage arose from internally generated sales, it would be more attractive.

3.4.3 Profitability Ratios Required for Analysis from Owner's Point of View

- (a) **Earnings per Share (EPS):** The profitability of a firm from the point of view of ordinary shareholders can be measured in terms of earnings per share basis. It is calculated as follows:

$$\text{Earnings per Share (EPS)} = \frac{\text{Net profit available to equity}}{\text{Number of equity shares}}$$

- (b) **Dividend per Share (DPS):** Earnings per share as stated above reflects the profitability of a firm per share; it does not reflect how much profit is paid as dividend and how much is retained by the business. Dividend per share ratio indicates the amount of profit distributed to equity shareholders per share. It is calculated as:

$$\text{Dividend per Share (DPS)} = \frac{\text{Total Dividend paid to equity}}{\text{Number of equity shares}}$$

- (c) **Dividend Pay-out Ratio (DP):** This ratio measures the dividend paid in relation to net earnings. It is determined to see to how much extent earnings per share have been retained by the management for the business. It is computed as:

$$\text{Dividend pay-out Ratio} = \frac{\text{Dividend per equity share}}{\text{Earning per Share}}$$

3.4.4 Profitability Ratios related to market/valuation/Investors

These ratios consider the market value of the company's shares in calculation. Frequently, share prices data are punched with the accounting data to generate new set of information. These are (a) Price-Earnings Ratio, (b) Dividend Yield, (c) Market Value/ Book Value per share, (d) Q Ratio.

- (a) **Price- Earnings Ratio (P/E Ratio):** The price earnings ratio indicates the **expectation of equity investors about the earnings of the firm**. It relates earnings to market price and is generally taken as a summary measure of growth potential of an investment, risk characteristics, shareholders orientation, corporate

$$\text{Price-Earnings per Share (P/E Ratio)} = \frac{\text{Market Price per}}{\text{Earning per Share}}$$

image and degree of liquidity. It is calculated as

Interpretation

It indicates the payback period to the investors or prospective investors. A higher P/E ratio could either mean that a company's stock is over-valued or the investors are expecting high growth rates in future.

- (b) **Dividend and Earning Yield:**

$$\begin{aligned} \text{Dividend Yield} &= \frac{\text{Dividend} \pm \text{Change in share price}}{\text{Initial share price}} \\ &= \frac{\text{Dividend per Share (DPS)}}{\text{Market Price per Share}} \end{aligned}$$

$$\text{Earnings Yield* or EP Ratio} = \frac{\text{Earnings per Share (EPS)}}{\text{Market Price per Share}}$$

*Also known as Earnings Price (EP) Ratio.

Interpretation

This ratio indicates return on investment; this may be on average investment or closing investment. Dividend (%) indicates return on paid up value of shares. But yield (%) is the indicator of true return in which share capital is taken at its market value.

- (c) **Market Value/ Book Value per Share (MV/BV):** It provides evaluation of how investors view the company's past and future performance.

Market Value/ Book Value per Share
(MV/BV) =

$$= \frac{\text{Average share price}}{\text{Net worth} \div \text{No. of equity shares}}$$

$$= \frac{\text{Closing share price}}{\text{Net worth} \div \text{No. of equity}}$$

Interpretation

This ratio indicates market response of the shareholders' investment. Undoubtedly, higher the ratio, better is the shareholders' position in terms of return and capital gains.

- (d) **Q Ratio:** This ratio is proposed by James Tobin, a ratio is defined as

$$\text{Q Ratio} = \frac{\text{Market Value of equity and}}{\text{Estimated replacement cost of assets}}$$

$$= \frac{\text{Market Value of a Company}}{\text{Estimated replacement cost of assets}}$$

Thus, this ratio represents the relationship between market valuation and intrinsic value. Equilibrium is when Q Ratio = 1 because when it is less than 1, it could mean that the stock is undervalued and when it is more than 1, it could mean that stock is overvalued.

Notes for calculating Ratios:

1. EBIT (Earnings before interest and taxes) = PBIT (Profit before interest and taxes),
EAT (Earnings after taxes) = PAT (Profit after taxes), EBT (Earnings before taxes) = PBT (Profit before taxes)
2. In absence of preference dividend PAT can be taken as earnings available to equity shareholders.
3. If information is available then average capital employed shall be taken while calculating ROCE.
4. Ratios shall be calculated based on requirement and availability of information and may deviate from original formulae. If required, assumptions should be given.
5. Numerator should be taken in correspondence with the denominator and vice-versa.



4. USERS AND OBJECTIVE OF FINANCIAL ANALYSIS - A BIRD'S EYE VIEW

Financial Statement analysis is useful to various shareholders to obtain the derived information about the firm.

S.No.	Users	Objectives	Ratios used in general
1.	Shareholders	Being owners of the organisation they are interested to know about profitability and growth of the organization	Mainly Profitability Ratios [In particular Earning per share (EPS), Dividend per share (DPS), Price Earnings (P/E), Dividend Payout ratio (DP)]

2.	Investors	They interested	are to	♦ Profitability Ratios
----	-----------	--------------------	-----------	------------------------

		know overall financial health of the organisation particularly future perspective of the organisations.	♦ Capital structure Ratios ♦ Solvency Ratios ♦ Turnover Ratios
3.	Lenders	They will keep an eye on the safety perspective of their money lent to the organisation	♦ Coverage Ratios ♦ Solvency Ratios ♦ Turnover Ratios ♦ Profitability Ratios
4.	Creditors	They are interested to know liability position of the organisation particularly in short term. Creditors would like to know whether the organisation will be able to pay the amount on due date.	♦ Liquidity Ratios ♦ Short term solvency Ratios/ Liquidity Ratios

5.	Employees	They will be interested to know the overall financial wealth of the organisation and compare it with competitor company.	♦ Liquidity Ratios ♦ Long terms solvency Ratios ♦ Profitability Ratios ♦ Return on investment
----	-----------	--	--

6.	Regulator / Government	They will analyse the financial statements to determine taxations and other details payable to the government.	♦ Profitability Ratios
7.	Managers		
	(a) Production Managers	They are interested to know about data regarding input output, production quantities etc.	♦ Input output Ratio ♦ Raw material consumption ratio.
	(b) Sales Managers	Data related to units sold for various years, other associated figures and predicted future sales figure will be an area of interest for them	♦ Turnover ratios (basically receivable turnover ratio) ♦ Expenses Ratios
	(c) Financial Manager	They are interested to know various ratios for their future predictions of	♦ Profitability Ratios (particularly related to Return on investment) ♦ Turnover ratios ♦ Capital Structure Ratios

		financial requirement.	
--	--	---------------------------	--

	(d) Chief Executive/ General Manager	They will try to assess the complete perspective of the company, starting from Sales, Finance, Inventory, Human resources, Production etc.	♦ All Ratios
8.	Different Industry	Finance Manager/ Analyst will calculate ratios of their company and compare it with Industry norms.	♦ Ratio related to 'call' and expenses per customer
	(a) Telecom		♦ Revenue and expenses per customer
	(b) Bank		♦ Loan to deposit Ratios ♦ Operating expenses and income ratios
	(c) Hotel		♦ Room occupancy ratio ♦ Bed occupancy Ratios
	(d) Transport		♦ Passenger-kilometre ♦ Operating cost-per passenger kilometre



5. APPLICATION OF RATIO ANALYSIS IN FINANCIAL DECISION MAKING

A popular technique of analysing the performance of a business concern is that of financial ratio analysis. As a tool of financial management, they are

of crucial significance.

The importance of ratio analysis lies in the fact that it presents facts on a comparative basis and enables drawing of inferences regarding the performance of a firm.

Ratio analysis is relevant in assessing the performance of a firm in respect of following aspects:

5.1 Financial Ratios for Evaluating Performance

(a) Liquidity Position: With the help of ratio analysis one can draw conclusions regarding liquidity position of a firm. The liquidity position of a firm would be satisfactory if it is able to meet its obligations when they become due. This ability is reflected in the liquidity ratios of a firm. The liquidity ratios are particularly useful in credit analysis by banks and other suppliers of short- term loans.

(b) Long-term Solvency: Ratio analysis is equally useful for assessing the long- term financial viability of a firm. This aspect of the financial position of a borrower is of concern to the long term creditors, security analysts and the present and potential owners of a business.

The long term solvency is measured by the leverage/capital structure and profitability ratios which focus on earning power and operating efficiency.

The leverage ratios, for instance, will indicate whether a firm has a reasonable proportion of various sources of finance or whether it is heavily loaded with debt in which case its solvency is exposed to serious strain.

Similarly, the various profitability ratios would reveal whether or not the firm is able to offer adequate return to its owners consistent with the risk involved.

(c) Operating Efficiency: Ratio analysis throws light on the degree of efficiency in the management and utilisation of its assets.

The various activity ratios measure this kind of operational efficiency. In fact, the solvency of a firm is, in the ultimate analysis, dependent upon the sales revenues generated by the use of its assets – total as well as its components.

- (d) Overall Profitability:** Unlike the outside parties which are interested in one aspect of the financial position of a firm, the management is constantly concerned about the overall profitability of the enterprise. That is, they are concerned about the ability of the firm to meet its short-term as well as long-term obligations to its creditors, to ensure a reasonable return to its owners and secure optimum utilisation of the assets of the firm. This is possible if an integrated view is taken and all the ratios are considered together.
- (e) Inter-firm Comparison:** Ratio analysis not only throws light on the financial position of a firm but also serves as a stepping stone to remedial measures. This is made possible due to inter-firm comparison/comparison with industry averages.

A single figure of particular ratio is meaningless unless it is related to some standard or norm. One of the popular techniques is to compare the ratios of a firm with the industry average. It should be reasonably expected that the performance of a firm should be in broad conformity with that of the industry to which it belongs.

An inter-firm comparison would demonstrate the relative position vis-a-vis its competitors. If the results are at variance either with the industry average or with those of the competitors, the firm can seek to identify the probable reasons and, in the light, take remedial measures.

Ratios not only perform post mortem of operations, but also serve as barometer for future. Ratios have predictor value and they are very helpful in forecasting and planning the business activities for a future.

Conclusions are drawn on the basis of the analysis obtained by using ratio analysis. The decisions affected may be whether to supply goods on credit to a concern, whether bank loans will be made available, etc.

- (f) Financial Ratios for Budgeting:** In this field ratios are able to provide a great deal of assistance. Budget is only an estimate of future activity based on past experience, in the making of which the relationship between different spheres of activities are invaluable.

It is usually possible to estimate budgeted figures using financial ratios.

Ratios also can be made use of for measuring actual performance with budgeted estimates. They indicate directions in which adjustments should be made either in the budget or in performance to bring them closer to each other.



6. LIMITATIONS OF FINANCIAL RATIOS

The limitations of financial ratios are listed below:

- (i) **Diversified product lines:** Many businesses operate a large number of divisions in quite different industries. In such cases ratios calculated on the basis of aggregate data cannot be used for inter-firm comparisons.
- (ii) **Financial data are badly distorted by inflation:** Historical cost values may be substantially different from true values. Such distortions of financial data are also carried in the financial ratios.
- (iii) **Seasonal factors:** It may also influence financial data.

Example: A company deals in cotton garments. It keeps a high inventory during October - January every year. For the rest of the year its inventory level becomes just 1/4th of the seasonal inventory level.

So, the liquidity ratios and inventory ratios will produce biased picture. Year end picture may not be the average picture of the business. Sometimes it is suggested to take monthly average inventory data instead of year end data to eliminate seasonal factors. But for external users it is difficult to get monthly inventory figures. (Even in some cases monthly inventory figures may not be available).

- (iv) **To give a good shape to the popularly used financial ratios (like current ratio, debt-equity ratios etc.):** The business may make some year-end adjustments. Such window dressing can change the character of financial ratios which would be different had there been no such change.
- (v) **Differences in accounting policies and accounting period:** It can make the accounting data of two firms non-comparable as also the accounting ratios.

- (vi) **No standard set of ratios against which a firm's ratios can be compared:** Sometimes a firm's ratios are compared with the industry average. But if a firm desires to be above the average, then industry average becomes a low standard. On the other hand, for a below average firm, industry averages become too high a standard to achieve.
- (vii) **Difficulty to generalise whether a particular ratio is good or bad:** For example, a low current ratio may be said 'bad' from the point of view of low liquidity, but a high current ratio may not be 'good' as this may result from inefficient working capital management.
- (viii) **Financial ratios are inter-related, not independent:** Viewed in isolation one ratio may highlight efficiency. But when considered as a set of ratios they may speak differently. Such interdependence among the ratios can be taken care of through multivariate analysis (analyzing the relationship between several variables simultaneously).

Financial ratios provide clues but not conclusions. These are tools only in the hands of experts because there is no standard ready-made interpretation of financial ratios.

7. FINANCIAL ANALYSIS

It may be of two types: - Horizontal and vertical.

Horizontal Analysis: When financial statement of one year are analysed and interpreted after comparing with another year or years, it is known as horizontal analysis. It can be based on the ratios derived from the financial information over the same time span.

Vertical Analysis: When financial statement of single year is analyzed then it is called vertical analysis. This analysis is useful in inter firm comparison. Every item of Profit and loss account is expressed as a percentage of gross sales, while every item on a balance sheet is expressed as a percentage of total assets held by the firm.

8. SUMMARY OF RATIOS

Another way of categorizing the ratios is being shown to you in a tabular form. A summary of the ratios has been tabulated as under:

Ratio	Formulae	Interpretation
Liquidity Ratio		
Current Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$	A simple measure that estimates whether the business can pay short term debts. Ideal ratio is 2.
Quick Ratio	$\frac{\text{Quick Assets}}{\text{Current Liabilities}}$	It measures the ability to meet current debt immediately. Ideal ratio is 1.
Cash Ratio	$\frac{\text{Cash and Bank balances} + \text{Marketable Securities}}{\text{Current Liabilities}}$	It measures absolute liquidity of the business.
Basic Defense Interval Ratio	$\frac{\text{Cash and Bank balances} + \text{Net Receivables} + \text{Marketable Securities}}{\text{Operating Expenses No. of days}}$	It measures the ability of the business to meet regular cash expenditures.
Net Working Capital	$\text{Current Assets} - \text{Current Liabilities}$	It is a measure of cash flow to determine the ability of business to survive financial crisis.
Capital Structure Ratio		
Equity Ratio	$\frac{\text{Shareholders' Equity}}{\text{Net Assets}}$	It indicates owner's fund in companies to total fund invested.
Debt Ratio	$\frac{\text{Total Debt}}{\text{Net Assets}}$	It is an indicator of use of outside funds.
Debt to equity Ratio	$\frac{\text{Total Debt}}{\text{Shareholders' Equity}}$	It indicates the composition of capital structure in terms of debt and equity.

Debt to Total Assets Ratio	$\frac{\text{Total Debt}}{\text{Total Assets}}$	It measures how much of total assets is financed by the debt.
Capital Gearing Ratio	$\frac{\text{Preference Share Capital} + \text{Debentures} + \text{Other Borrowed funds}}{\text{Equity Share Capital} + \text{Reserves \& Surplus} - \text{Losses}}$	It shows the proportion of fixed interest bearing capital to equity shareholders' fund. It also signifies the advantage of financial leverage to the equity shareholder.
Proprietary Ratio	$\frac{\text{Proprietary Fund}}{\text{Total Assets}}$	It measures the proportion of total assets financed by shareholders.
Coverage Ratios		
Debt Service Coverage Ratio (DSCR)	$\frac{\text{Earnings available for debt services}}{\text{Interest} + \text{Instalments}}$	It measures the ability to meet the commitment of various debt services like interest, instalment etc. Ideal ratio is 2.
Interest Coverage Ratio	$\frac{\text{EBIT}}{\text{Interest}}$	It measures the ability of the business to meet interest obligations. Ideal ratio is > 1.
Preference Dividend Coverage Ratio	$\frac{\text{Net Profit / Earning after taxes (EAT)}}{\text{Preference dividend liability}}$	It measures the ability to pay the preference shareholders' dividend. Ideal ratio is > 1.
Fixed Charges Coverage Ratio	$\frac{\text{EBIT} + \text{Depreciation}}{\text{Interest} + \text{Repayment of loan}}$	This ratio shows how many times the cash flow before interest and taxes covers all fixed financing charges. The ideal ratio is > 1.
Activity Ratio/ Efficiency Ratio/ Performance Ratio/ Turnover Ratio		

Total Asset Turnover Ratio	$\frac{\text{Sales / Cost of Goods Sold}}{\text{Average Total Assets}}$	A measure of total asset utilisation. It helps to answer the question - What
---	---	---

		sales are being generated by each rupee's worth of assets invested in the business?
Fixed Assets Turnover Ratio	$\frac{\text{Sales / Cost of Goods Sold}}{\text{Fixed Assets}}$	This ratio is about fixed asset capacity. A reducing sales or profit being generated from each rupee invested in fixed assets may indicate overcapacity or poorer-performing equipment.
Capital Turnover Ratio	$\frac{\text{Sales / Cost of Goods Sold}}{\text{Net Assets}}$	This indicates the firm's ability to generate sales per rupee of long term investment.
Working Capital Turnover Ratio	$\frac{\text{Sales / COGS}}{\text{Working Capital}}$	It measures the efficiency of the firm to use working capital.
Inventory Turnover Ratio	$\frac{\text{COGS / Sales}}{\text{Average Inventory}}$	It measures the efficiency of the firm to manage its inventory.
Debtors Turnover Ratio	$\frac{\text{Credit Sales}}{\text{Average Accounts Receivable}}$	It measures the efficiency at which firm is managing its receivables.
Receivables (Debtors') Velocity	$\frac{\text{Average Accounts Receivables}}{\text{Average Daily Credit Sales}}$	It measures the velocity of collection of receivables.
Payables Turnover Ratio	$\frac{\text{Annual Net Credit Purchases}}{\text{Average Accounts Payables}}$	It measures how fast a company makes payment to its creditors.
Payables Velocity	$\frac{\text{Average Accounts Payable}}{\text{Average Daily Credit Purchases}}$	It measures the velocity of payment of payables.

Profitability Ratios based on Sales		
Gross Profit Ratio	$\frac{\text{Gross Profit}}{\text{Sales}} \times 100$	This ratio tells us something about the business's ability consistently to control its production costs or to manage the margins it makes on products it buys and sells.
Net Profit Ratio	$\frac{\text{Net Profit}}{\text{Sales}} \times 100$	It measures the relationship between net profit and sales of the business.
Operating Profit Ratio	$\frac{\text{Operating Profit}}{\text{Sales}} \times 100$	It measures operating performance of business.
Expenses Ratio		
Cost of Goods Sold (COGS) Ratio	$\frac{\text{COGS}}{\text{Sales}} \times 100$	It measures portion of a particular expenses in comparison to sales.
Operating Expenses Ratio	$\frac{\text{Administrative exp. + (Selling \& Distribution Overhead)}}{\text{Sales}} \times 100$	
Operating Ratio	$\frac{\text{COGS + Operating expenses}}{\text{Sales}} \times 100$	
Financial Expenses Ratio	$\frac{\text{Financial expenses}}{\text{Sales}} \times 100$	
Profitability Ratios related to Overall Return on Assets/ Investments		
Return on Investment (ROI)	$\frac{\text{Return / Profit / Earnings}}{\text{Investments}} \times 100$	It measures overall return of the business on investment/ equity funds/capital employed/ assets.

Return on Assets (ROA)	$\frac{\text{Net Profit after taxes}}{\text{Average total assets}}$	It measures net profit per rupee of average total assets/average tangible assets/average fixed assets.
------------------------	---	--

Return on Capital Employed ROCE (Pre-tax)	$\frac{\text{EBIT}}{\text{Capital Employed}} \times 100$	It measures overall earnings (either pre-tax or post tax) on total capital employed. It indicates earnings available to equity shareholders in comparison to equity shareholders' net worth.
Return on Capital Employed ROCE (Post-tax)	$\frac{\text{EBIT (1-t)}}{\text{Capital Employed}} \times 100$	
Return on Equity (ROE)	$\frac{\text{Net Profit after taxes - Preference dividend (if any)}}{\text{Net worth of Equity shareholders' fund}} \times 100$	

Profitability Ratios Required for Analysis from Owner's Point of View

Earnings per Share (EPS)	$\frac{\text{Net profit available to equity shareholders}}{\text{Number of equity shares outstanding}}$	EPS measures the overall profit generated for each share in existence over a particular period.
Dividend per Share (DPS)	$\frac{\text{Dividend paid to equity shareholders}}{\text{Number of equity shares outstanding}}$	Proportion of profit distributed per equity share.
Dividend payout Ratio (DP)	$\frac{\text{Dividend per equity share}}{\text{Earning per Share (EPS)}}$	It shows % of EPS paid as dividend and retained earnings.

Profitability Ratios related to market/ valuation/ Investors

Price-Earnings per Share (P/E Ratio)	$\frac{\text{Market Price per Share (MPS)}}{\text{Earning per Share (EPS)}}$	At any time, the P/E ratio is an indication of how highly the market "rates" or "values" a business. A P/E ratio is best viewed in the context of a sector or market average to get a feel for relative value and stock market pricing.
--------------------------------------	--	---

Dividend Yield	$\frac{\text{Dividend} \pm \text{Change in share price}}{100 \text{ Initial share price}}$	It measures dividend paid based on market price of shares.
----------------	--	--

	$\frac{\text{OR Dividend per Share (DPS)}}{\text{Market Price per Share}} \times 100$ (MPS)	
Earnings Yield	$\frac{\text{Earnings per Share (EPS)}}{\text{Market Price per Share}} \times 100$ (MPS)	It is the relationship of earning per share and market value of shares.
Market Value /Book Value per Share	$\frac{\text{Market value per share}}{\text{Book value per share}}$	It indicates market response of the shareholders' investment.
Q Ratio	$\frac{\text{Market Value of equity and liabilities}}{\text{Estimated replacement cost of assets}}$	It measures market value of equity as well as debt in comparison to all assets at their replacement cost.

Students may note that now a company is also required to disclose the following ratios in the notes to accounts while preparing Financial Statements:

- (a) Current Ratio,
- (b) Debt-Equity Ratio,
- (c) Debt Service Coverage Ratio,
- (d) Return on Equity Ratio,
- (e) Inventory turnover ratio,
- (f) Trade Receivables turnover ratio,
- (g) Trade payables turnover ratio,
- (h) Net capital turnover ratio,
- (i) Net profit ratio,
- (j) Return on Capital employed,
- (k) Return on investment.

ILLUSTRATION 1

In a meeting held at Solan towards the end of 2021-22, the Directors of HPCL Ltd. have taken a decision to diversify. At present HPCL Ltd. sells all finished goods from its own warehouse. The company issued debentures on 01.04.2022 and purchased fixed assets on the same day. The purchase prices have remained stable during the concerned period. Following information is provided to you:

INCOME STATEMENT

Particulars	2021-22 (₹)		2022-23 (₹)	
Cash Sales	30,000		32,000	
Credit Sales	2,70,000	3,00,000	3,42,000	3,74,000
Less: Cost of goods sold		2,36,000		2,98,000
Gross profit		64,000		76,000
Less: Operating Expenses:				
Warehousing	13,000		14,000	
Transport	6,000		10,000	
Administrative	19,000		19,000	
Selling	11,000	49,000	14,000	57,000
Net Profit		15,000		19,000

BALANCE SHEET

Assets & Liabilities	2021-22 (₹)		2022-23 (₹)	
Fixed Assets (Net Block)	-	30,000	-	40,000
Receivables	50,000		82,000	
Cash at Bank	10,000		7,000	
Stock	60,000		94,000	
Total Current Assets (CA)	1,20,000		1,83,000	
Payables	50,000		76,000	
Total Current Liabilities (CL)	50,000		76,000	
Working Capital (CA - CL)		70,000		1,07,000

<i>Net Assets</i>		<i>1,00,000</i>		<i>1,47,00 0</i>
-------------------	--	-----------------	--	----------------------

Represented by:				
Share Capital		75,000		75,000
Reserve and Surplus		25,000		42,000
Debentures		~		30,000
		1,00,000		1,47,000

You are required to CALCULATE the following ratios for the years 2021-22 and 2022-23:

- (i) Gross Profit Ratio
- (ii) Operating Expenses to Sales Ratio
- (iii) Operating Profit Ratio
- (iv) Capital Turnover Ratio
- (v) Stock Turnover Ratio
- (vi) Net Profit to Net Worth Ratio
- (vii) Receivables Collection Period

Ratio relating to capital employed should be based on the capital at the end of the year. Give the reasons for change in the ratios for 2 years. Assume opening stock of ₹ 40,000 for the year 2021-22. Ignore Taxation.

SOLUTION

Computation of Ratios		
Ratio	2021-22 (₹)	2022-23 (₹)
1. Gross profit ratio (Gross profit/sales)	$\frac{64,000}{3,00,000} \times 100$ =21.3%	$\frac{76,000}{3,74,000} \times 100$ =20.3%
2. Operating expense to sales ratio (Operating exp/ Total sales)	$\frac{49,000}{3,00,000} \times 100$ =16.3%	$\frac{57,000}{3,74,000} \times 100$ =15.2%
3. Operating profit ratio (Operating profit/ Total sales)	$\frac{15,000}{3,00,000} \times 100$ =5%	$\frac{19,000}{3,74,000} \times 100$ =5.08%

4. Capital turnover ratio (Sales / capital employed)	$\frac{3,00,000}{1,00,000} = 3$	$\frac{3,74,000}{1,47,000} = 2.54$
5. Stock turnover ratio (COGS/ Average stock) (Refer to W.N. 1)	$\frac{2,36,000}{50,000} = 4.72$	$\frac{2,98,000}{77,000} = 3.87$
6. Net Profit to Net worth ratio (Net profit / Net worth)	$\frac{15,000 \times 100}{1,00,000} = 15\%$	$\frac{19,000 \times 100}{1,17,000} = 16.24\%$
7. Receivables collection period (Average receivables/ Average daily credit sales) (Refer to W.N. 2)	$\frac{50,000}{739.73} = 67.6$ days	$\frac{82,000}{936.99} = 87.5$ days
Working notes (W.N.):		
1. Average Stock = (opening stock + closing stock)/2	$(40,000 + 60,000)/2 = 50,000$	$(60,000 + 94,000)/2 = 77,000$
2. Average daily sales = Credit sales / 365	$\frac{2,70,000}{365} = 739.73$	$\frac{3,42,000}{365} = 936.99$

Analysis: The decline in the Gross profit ratio could be either due to a reduction in the selling price or increase in the direct expenses (since the purchase price has remained the same). In this case, cost of goods sold have increased more than proportion of increment in sales & hence impacting gross profit ratio.

Similarly, there is a decline in the ratio of operating expenses to sales. Further analysis reveals that in comparison to increase in sales, there has a lesser proportionate increase in operating expenses. As a result, even the operating profit ratio has remained the same approximately in spite of a decline in the Gross profit ratio.

The company has not been able to deploy its capital efficiently. This is indicated by a decline in the Capital turnover ratio from 3 to 2.54 times.

The decline in stock turnover ratio implies that the company has increased its investment in stock. Net Profit to Net worth ratio has increased indicating that the company's Net worth or Shareholders' capital is

efficient in generating profits.

The increase in the Receivables collection period indicates that the company has become liberal in extending credit on sales. There is a corresponding increase in the receivables also due to such credit policy.

ILLUSTRATION 2

Following is the abridged Balance Sheet of Alpha Ltd.:

Liabilities	`	Assets	`	`
Share Capital	1,00,000	Land and Buildings		80,000
Profit and Loss Account	17,000	Plant and Machineries	50,000	
Current Liabilities	40,000	Less: Depreciation	15,000	35,000
				1,15,000
		Stock	21,000	
		Receivables	20,000	
		Bank	1,000	42,000
Total	1,57,000	Total		1,57,000

With the help of the additional information furnished below, you are required to PREPARE Trading and Profit & Loss Account and Balance Sheet as at 31st March, 2023:

- (i) The company went in for re-organisation of capital structure, with share capital remaining the same as follows:

Share capital	50%
Other Shareholders' funds	15%
5% Debentures	10%
Current Liabilities	25%

Debentures were issued on 1st April, interest being paid annually on 31st March.

- (ii) Land and Buildings remained unchanged. Additional plant and machinery has been bought and a further ` 5,000 depreciation was written off.

(The total fixed assets then constituted 60% of total fixed and current assets.)

- (iii) Working capital ratio was 8 : 5.*
- (iv) Quick assets ratio was 1 : 1.*
- (v) The receivables (four-fifth of the quick assets) to sales ratio revealed a credit period of 2 months. There were no cash sales.*
- (vi) Return on net worth was 10%.*
- (vii) Gross profit was at the rate of 15% of selling price.*
- (viii) Stock turnover was eight times for the year. Ignore Taxation.*

SOLUTION

Particulars	%	(`)
Share capital (given to be same)	50%	1,00,000
Other shareholders funds	15%	30,000
5% Debentures	10%	20,000
Current Liabilities	25%	50,000
Total (1,00,000 / 50%)	100%	2,00,000

Calculation of Assets

$$\begin{aligned}\text{Total liabilities} &= \text{Total Assets} \\ ` 2,00,000 &= \text{Total Assets} \\ \text{Fixed Assets} &= 60\% \text{ of total fixed assets and current assets} \\ &= ` 2,00,000 \times \frac{60}{100} = ` 1,20,000 \\ \text{Current Assets} &= \text{Total Assets} - \text{Fixed Assets} \\ &= ` 2,00,000 - ` 1,20,000 = ` 80,000\end{aligned}$$

Calculation of additions to Plant & Machinery

	₹
Total fixed assets	1,20,000
Less: Land & Buildings	80,000
Plant and Machinery (after providing depreciation)	40,000
Less: Existing Plant & Machinery (after extra depreciation of ₹ 5,000) i.e. 50,000 - 20,000	30,000
Addition to the Plant & Machinery	10,000

Calculation of stock

$$\begin{aligned} \text{Quick ratio:} &= \frac{\text{Current assets} - \text{stock}}{\text{Current liabilities}} \\ &= \frac{₹ 80,000 - \text{stock}}{₹ 50,000} = 1 \end{aligned}$$

$$₹ 50,000 = ₹ 80,000 - \text{Stock}$$

$$\begin{aligned} \text{Stock} &= ₹ 80,000 - ₹ 50,000 \\ &= ₹ 30,000 \end{aligned}$$

$$\begin{aligned} \text{Receivables} &= \frac{4}{5}^{\text{th}} \text{ of quick assets} \\ &= \frac{(₹ 80,000 - ₹ 30,000)}{\frac{4}{5}} \\ &= ₹ 40,000 \end{aligned}$$

$$\begin{aligned} \text{Receivables turnover} &= \frac{\text{Receivables}}{\text{Credit Sales}} \times 12 \text{ Months} = 2 \text{ months} \\ &= \frac{40,000}{\text{Credit Sales}} \times 12 = 2 \text{ months} \end{aligned}$$

$$2 \times \text{credit sales} = 4,80,000$$

$$\begin{aligned} \text{Credit sales} &= 4,80,000 / 2 \\ &= ₹ 2,40,000 = \text{Total Sales (As there were no cash sales)} \end{aligned}$$

$$\text{Gross profit} = 15\% \text{ of sales} = ₹ 2,40,000 \times \frac{15}{100} = ₹ 36,000$$

Return on net worth (net profit)

Net worth = ₹ 1,00,000 + ₹ 30,000

= ₹ 1,30,000

Net profit = ₹ 1,30,000 $\times$ 10/100 = ₹

13,000
Debenture interest = ₹

20,000 $\times$ 5/100 = ₹ 1,000

Projected profit and loss account for the year ended 31st March, 2023

Particulars	₹	Particulars	₹
To cost of goods sold	2,04,000	By sales	2,40,000
To gross profit	36,000		0
	2,40,000		2,40,000
To debenture interest	1,000	By gross profit	36,000
To administration and other expenses (bal. fig.)	22,000		
To net profit	13,000		
	36,000		36,000

Projected Balance Sheet as at 31st March, 2023

Liabilities	₹	Assets		₹
Share capital	1,00,000	Fixed assets:		
Profit and loss A/c (17,000+13,000)	30,000	Land & buildings		80,000
5% Debentures	20,000	Plant & machinery	60,000	
Current liabilities	50,000	Less: Depreciation	20,000	40,000
		Current assets		
		Stock	30,000	
		Receivables	40,000	
		Bank	10,000	80,000
	2,00,000			2,00,000

ILLUSTRATION 3

X Co. has made plans for the next year. It is estimated that the company will employ total assets of ₹ 8,00,000; 50 per cent of the assets being financed by borrowed capital at an interest cost of 8 per cent per year. The direct costs for the year are estimated at ₹ 4,80,000 and all other operating expenses are estimated at ₹ 80,000. The goods will be sold to customers at 150 per cent of the direct costs. Tax rate is assumed to be 50 per cent.

You are required to CALCULATE: (i) Operating profit margin (before tax); (ii) net profit margin (after tax); (iii) return on assets (on operating profit after tax); (iv) asset turnover and (v) return on owners' equity.

SOLUTION

The net profit is calculated as follows:

Particulars	₹
Sales (150% of ₹ 4,80,000)	7,20,000
Direct costs	(4,80,000)
Gross profit	2,40,000
Operating expenses	(80,000)
Profit before Interest and Tax (EBIT)	1,60,000
Interest charges (8% of ₹ 4,00,000)	(32,000)
Profit before taxes	1,28,000
Taxes (@ 50%)	(64,000)
Net profit after taxes	64,000

$$(i) \text{ Operating profit margin} = \frac{\text{EBIT}}{\text{Sales}} = \frac{1,60,000}{7,20,000} = 0.2222 \text{ or } 22.22\%$$

$$(ii) \text{ Net profit margin} = \frac{\text{Net Profit after taxes}}{\text{Sales}} = \frac{64,000}{7,20,000} = 0.089 \text{ or } 8.9\%$$

$$(iii) \text{ Return on assets} = \frac{\text{EBIT} (1 - T)}{\text{Assets}} = \frac{1,60,000(1 - 0.5)}{8,00,000}$$

= 0.10 or 10%

$$\begin{aligned}
 \text{(iv) Asset turnover} &= \frac{\text{Sales}}{\text{Assets}} = \frac{7,20,000}{8,00,000} = 0.9 \text{ times} \\
 \text{(v) Return on equity} &= \frac{\text{Net Profit after taxes}}{\text{Owners' equity}} = \frac{64,000}{50\% \text{ of } 8,00,000} \\
 &= \frac{64,000}{4,00,000} = 0.16 \text{ or } 16\%
 \end{aligned}$$

ILLUSTRATION 4

From the following ratios and information given below, PREPARE Trading Account, Profit and Loss Account and Balance Sheet of Aebece Company:

Fixed Assets	₹ 40,00,000
Closing Stock	₹ 4,00,000
Stock turnover ratio	10
Gross profit ratio	25 percent
Net profit ratio	20 percent
Net profit to capital	1/5
Capital to total liabilities	1/2
Fixed assets to capital	5/4
Fixed assets/Total current assets	5/7

SOLUTION

Workings:

$$\begin{aligned}
 \text{(i) } \frac{\text{Fixed Assets}}{\text{Total Current Assets}} &= \frac{5}{7} \\
 \text{Or, Total Current Assets} &= \frac{₹ 40,00,000 \times 7}{5} = ₹ 56,00,000 \\
 \text{(ii) } \frac{\text{Fixed Assets}}{\text{Capital}} &= \frac{5}{4} \\
 \text{Or, Capital} &= \frac{₹ 40,00,000 \times 4}{5} = ₹ 32,00,000
 \end{aligned}$$

$$(iii) \frac{\text{Capital}}{\text{Total Liabilities}} = \frac{1}{2}$$

Total Liabilities* 2

$$\text{Or, Total liabilities} = ₹ 32,00,000 \times 2 = ₹ 64,00,000$$

*It is assumed that total liabilities do not include capital.

$$(iv) \frac{\text{Net Profit}}{\text{Capital}} = \frac{1}{5}$$

Capital 5

$$\text{Or, Net Profit} = ₹ 32,00,000 \times \frac{1}{5} = ₹ 6,40,000$$

$$(v) \frac{\text{Net Profit}}{\text{Sales}} = \frac{1}{5}$$

Sales 5

$$\text{Or, Sales} = ₹ 6,40,000 \times 5 = ₹ 32,00,000$$

$$(vi) \text{Gross Profit} = 25\% \text{ of } ₹ 32,00,000 = ₹ 8,00,000$$

$$(vii) \text{Stock Turnover} = \frac{\text{Cost of Goods Sold (i.e. Sales - Gross profit)}}{\text{Average Stock}} = 10$$

$$= \frac{₹ 32,00,000 - ₹ 8,00,000}{\text{Average Stock}} = 10$$

$$\text{Or, Average Stock} = ₹ 2,40,000$$

$$\text{Or, } \frac{\text{Opening Stock} + ₹ 4,00,000}{2} = ₹ 2,40,000$$

$$\text{Or, Opening Stock} = ₹ 80,000$$

Trading Account

Particulars	(₹)	Particulars	(₹)
To Opening Stock	80,000	By Sales	32,00,000
To Manufacturing exp./ Purchase	27,20,000		
(Balancing figure)			
To Gross Profit b/d	8,00,000	By Closing Stock	4,00,000
	36,00,000		36,00,000

Profit and Loss Account

Particulars	(`)	Particulars	(`)
To Operating Expenses (Balancing figure)	1,60,000	By Gross Profit c/d	8,00,000
To Net Profit	6,40,000		
	8,00,000		8,00,000

Balance Sheet

Capital and Liabilities	(`)	Assets	(`)
Capital	32,00,000	Fixed Assets	40,00,000
Liabilities	64,00,000	Current Assets:	
		Closing Stock	4,00,000
		Other	
		Current Assets (Bal. figure)	52,00,000
	96,00,000		96,00,000

ILLUSTRATION 5

ABC Company sells plumbing fixtures on terms of 2/10, net 30. Its financial statements over the last 3 years are as follows:

Particulars	2020-21	2021-22	2022-23
	`	`	`
Cash	30,000	20,000	5,000
Accounts receivable	2,00,000	2,60,000	2,90,000
Inventory	4,00,000	4,80,000	6,00,000
	6,30,000	7,60,000	8,95,000
Net fixed assets	8,00,000	8,00,000	8,00,000
	14,30,000	15,60,000	16,95,000

	2020-21	2021-22	2022-23
<i>Accounts payable</i>	2,30,000	3,00,000	3,80,000
<i>Accruals</i>	2,00,000	2,10,000	2,25,000
<i>Bank loan (short-term)</i>	1,00,000	1,00,000	1,40,000
	5,30,000	6,10,000	7,45,000
<i>Long-term debt</i>	3,00,000	3,00,000	3,00,000
<i>Common stock</i>	1,00,000	1,00,000	1,00,000
<i>Retained earnings</i>	5,00,000	5,50,000	5,50,000
	14,30,000	15,60,000	16,95,000
<i>Sales</i>	40,00,000	43,00,000	38,00,000
<i>Cost of goods sold</i>	32,00,000	36,00,000	33,00,000
<i>Net profit</i>	3,00,000	2,00,000	1,00,000

Considering opening balance of Accounts Receivable and Inventory as 2,00,000 and 4,00,000 respectively as on 01.04.2020, ANALYSE the company's financial condition and performance over the last 3 years. Are there any problems?

SOLUTION

Ratios	2020-21	2021-22	2022-23
Current ratio (Current Assets / Current Liabilities)	1.19 $\frac{\text{₹ } 6,30,000}{\text{₹ } 5,30,000}$ 	1.25 $\frac{\text{₹ } 7,60,000}{\text{₹ } 6,10,000}$ 	1.20 $\frac{\text{₹ } 8,95,000}{\text{₹ } 7,45,000}$
Acid-test ratio (Quick Assets / Current Liabilities)	0.43 $\frac{\text{₹ } 2,30,000}{\text{₹ } 5,30,000}$ 	0.46 $\frac{\text{₹ } 2,80,000}{\text{₹ } 6,10,000}$ 	0.40 $\frac{\text{₹ } 2,95,000}{\text{₹ } 7,45,000}$

Receivables turnover ratio (Sales/ Average Receivables) (Refer Working Notes)	20 $\frac{40,00,000}{2,00,000}$	18.70 $\frac{43,00,000}{2,30,000}$	13.82 $\frac{38,00,000}{2,75,000}$
Average collection period (365 / Receivables turnover ratio)	18.25 (365/20)	19.52 (365/18.70)	26.41 (365/13.82)
Inventory turnover ratio (COGS / Average Inventory) (Refer Working Notes)	8 $\frac{32,00,000}{4,00,000}$	8.18 $\frac{36,00,000}{4,40,000}$	6.11 $\frac{33,00,000}{5,40,000}$
Total debt to net worth (Short term + Long term Debt) / (Common stock + Retained earnings)	1.38 $\frac{8,30,000}{6,00,000}$	1.40 $\frac{9,10,000}{6,50,000}$	1.61 $\frac{10,45,000}{6,50,000}$
Long-term debt to total capitalization	0.33 $\frac{3,00,000}{9,00,000}$	0.32 $\frac{3,00,000}{9,50,000}$	0.32 $\frac{3,00,000}{9,50,000}$
Gross profit margin (Gross Profit / Sales) {Gross profit = Sales - Cost of Goods sold}	0.20 $\frac{8,00,000}{40,00,000}$	0.16 $\frac{7,00,000}{43,00,000}$	0.13 $\frac{5,00,000}{38,00,000}$
Net profit margin (Net Profit / Sales)	0.075 $\frac{3,00,000}{40,00,000}$	0.047 $\frac{2,00,000}{43,00,000}$	0.026 $\frac{1,00,000}{38,00,000}$
Total Asset turnover (Sales / Total Assets)	2.80 $\frac{40,00,000}{14,30,000}$	2.76 $\frac{43,00,000}{15,60,000}$	2.24 $\frac{38,00,000}{16,95,000}$

Return on assets (Net profit/ Total Assets)	0.21 $\frac{\text{₹ } 3,00,000}{\text{₹ } 14,30,000}$	0.13 $\frac{\text{₹ } 2,00,000}{\text{₹ } 15,60,000}$	0.06 $\frac{\text{₹ } 1,00,000}{\text{₹ } 16,95,000}$
Working Notes			
Average receivables {(Opening + closing)/2}	$\left(\frac{\text{₹ } 2,00,000 + \text{₹ } 2,00,000}{2} \right)$ $= \text{₹ } 2,00,000$	$\left(\frac{\text{₹ } 2,00,000 + \text{₹ } 2,60,000}{2} \right)$ $= \text{₹ } 2,30,000$	$\left(\frac{\text{₹ } 2,60,000 + \text{₹ } 2,90,000}{2} \right)$ $= \text{₹ } 2,75,000$
Average Inventory {(Opening + closing)/2}	$\left(\frac{\text{₹ } 4,00,000 + \text{₹ } 4,00,000}{2} \right)$ $= \text{₹ } 4,00,000$	$\left(\frac{\text{₹ } 4,00,000 + \text{₹ } 4,80,000}{2} \right)$ $= \text{₹ } 4,40,000$	$\left(\frac{\text{₹ } 4,80,000 + \text{₹ } 6,00,000}{2} \right)$ $= \text{₹ } 5,40,000$

Analysis: The current ratio and quick ratio are less than the ideal ratio (2:1 and 1:1 respectively) indicating that the company is not having enough resources to meet its current obligations.

Receivables are growing slower, although the average collection period is still very reasonable relative to the terms given. Inventory turnover is slowing as well, indicating a relative build-up in inventories. The increase in receivables and inventories, coupled with the fact that net worth has increased very little, has resulted in the total debt-to-net worth ratio increasing to what would have to be regarded on an absolute basis as a high level.

Long-term debt to total capitalization has not changed relatively coupled with the fact that retained earnings of only ₹ 50,000 is made in year 2019-20, and there is no issuance of new long-term debt in year 2019-20 and 2020-21.

Both the gross profit and net profit margins have declined substantially. The relationship between the two suggests that the company has incurred more relative expenses. The build-up in inventories and receivables has resulted in a decline in the asset turnover ratio, and this, coupled with the decline in profitability, has resulted in a sharp decrease in the return on assets ratio.

ILLUSTRATION 6

Following information are available for Navya Ltd. along with various ratios relevant to the particular industry it belongs to. APPRAISE your comments on strength and weakness of Navya Ltd. comparing its ratios with the given industry norms.

Navya Ltd.**Balance Sheet as at 31.3.2023**

Liabilities	(₹)	Assets	(₹)
Equity Share Capital	48,00,000	Fixed Assets	24,20,000
10% Debentures	9,20,000	Cash	8,80,000
Sundry Creditors	6,60,000	Sundry debtors	11,00,000
Bills Payable	8,80,000	Stock	33,00,000
Other current Liabilities	4,40,000		-
Total	77,00,000	Total	77,00,000

**Statement of
Profitability For the year**

Particulars	(₹)	(₹)
Sales		1,10,00,000
Less: Cost of goods sold:		
Material	41,80,000	
Wages	26,40,000	
Factory Overhead	12,98,000	81,18,000
Gross Profit		28,82,000
Less: Selling and Distribution Cost	11,00,000	
Administrative Cost	12,28,000	23,28,000
Earnings before Interest and Taxes		5,54,000
Less: Interest Charges		92,000
Earning before Tax		4,62,000
Less: Taxes @ 50%		2,31,000
Net Profit (PAT)		2,31,000

Industry Norms

Ratios	Norm
Current Ratio	2.5
Receivables Turnover Ratio	8.0
Inventory Turnover Ratio (based on Sales)	9.0
Total Assets Turnover Ratio	2.0
Net Profit Ratio	3.5%
Return on Total Assets (on EBIT)	7.0%
Return on Net worth (Based on Net profit)	10.5%
Total Debt/Total Assets	60.0%

SOLUTION

Ratios	Navya Ltd.	Industry Norms
1. Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$	$\frac{₹52,80,000}{₹19,80,000} = 2.67$	2.50
2. Receivable Turnover Ratio = $\frac{\text{Sales}}{\text{Debtors}}$	$\frac{₹1,10,00,000}{₹11,00,000} = 10.0$	8.00
3. Inventory turnover ratio = $\frac{\text{Sales}}{\text{Stock}}$	$\frac{₹1,10,00,000}{₹33,00,000} = 3.33$	9.00
4. Total Asset Turnover ratio = $\frac{\text{Sales}}{\text{Total Assets}}$	$\frac{₹1,10,00,000}{₹77,00,000} = 1.43$	2.00
5. Net Profit Ratio = $\frac{\text{Net Profit}}{\text{Sales}}$	$\frac{₹2,31,000}{₹1,10,00,000} = 2.10\%$	3.50%
6. Return on Total Asset = $\frac{\text{EBIT}}{\text{Total Assets}}$	$\frac{₹5,54,000}{₹77,00,000} = 7.19\%$	7%
7. Return on Net worth = $\frac{\text{Net Profit}}{\text{Net Worth}}$	$\frac{₹2,31,000}{₹48,00,000} = 4.81\%$	10.5%

8.	$\frac{\text{Total Debt Total}}{\text{Assets}}$	$\frac{29,00,000}{77,00,000} = 37.66\%$	60%
----	---	---	-----

Comments:

1. The position of Navya Ltd. is better than the industry norm with respect to Current Ratio and Receivables Turnover Ratio.
2. However, the Inventory turnover ratio and Total Asset Turnover ratio is poor comparing to industry norm indicating that company is inefficient to utilize its inventory and assets.
3. The firm also has its net profit ratio and return on net worth ratio much lower than the industry norm.
4. Total debt to total assets ratio is lower than the industry standard which suggests that the firm is less levered by debt and more by equity resulting in less risky company.

SUMMARY

- ♦ **Financial Analysis and its Tools:** For the purpose of obtaining the material and relevant information necessary for ascertaining the financial strengths and weaknesses of an enterprise, it is necessary to analyze the data depicted in the financial statement. The financial manager has certain analytical tools which help in financial analysis and planning. The main tools are Ratio Analysis and Cash Flow Analysis.
- ♦ **Ratio Analysis:** The ratio analysis is based on the fact that a single accounting figure by itself may not communicate any meaningful information but when expressed as a relative to some other figure, it may definitely provide some significant information. Ratio analysis is not just comparing different numbers from the balance sheet, income statement, and cash flow statement. It is comparing the number against previous years, other companies, the industry, or even the economy in general for the purpose of financial analysis.
- ♦ **Type of Ratios and Importance of Ratios Analysis:** The ratios can be classified into following four broad categories:
 - (i) Liquidity Ratios
 - (ii) Capital Structure/Leverage Ratios
 - (iii) Activity Ratios

(iv) Profitability Ratios

- ♦ A popular technique of analyzing the performance of a business concern is that of financial ratio analysis. As a tool of financial management, they are of crucial significance. The importance of ratio analysis lies in the fact that it presents facts on a comparative basis and enables drawing of inferences regarding the performance of a firm.
- ♦ Ratio analysis is relevant in assessing the performance of a firm in respect of following aspects:
 - I Liquidity Position
 - II Long-term Solvency
 - III Operating Efficiency
 - IV Overall Profitability
 - V Inter-firm Comparison
 - VI Financial Ratios for Supporting Budgeting

TEST YOUR KNOWLEDGE

Multiple Choice Questions (MCQs)

1. *Ratio of Net sales to Net working capital is a:*
 - (a) *Profitability ratio*
 - (b) *Liquidity ratio*
 - (c) *Current ratio*
 - (d) *Working capital turnover ratio*
2. *Long-term solvency is indicated by:*
 - (a) *Debt/equity ratio*
 - (b) *Current Ratio*
 - (c) *Operating ratio*
 - (d) *Net profit ratio*
3. *Ratio of net profit before interest and tax to sales is:*

- (a) *Gross profit ratio*
 - (b) *Net profit ratio*
 - (c) *Operating profit ratio*
 - (d) *Interest coverage ratio.*
4. *Observing changes in the financial variables across the years is:*
- (a) *Vertical analysis*
 - (b) *Horizontal Analysis*
 - (c) *Peer-firm Analysis*
 - (d) *Industry Analysis.*
5. *The Receivable-Turnover ratio helps management to:*
- (a) *Managing resources*
 - (b) *Managing inventory*
 - (c) *Managing customer relationship*
 - (d) *Managing working capital*
6. *Which of the following is a liquidity ratio?*
- (a) *Equity ratio*
 - (b) *Proprietary ratio*
 - (c) *Net Working Capital*
 - (d) *Capital Gearing ratio*
7. *Which of the following is not a part of Quick Assets?*
- (a) *Disposable investments*
 - (b) *Receivables*
 - (c) *Cash and Cash equivalents*
 - (d) *Prepaid expenses*

8. *Capital Gearing ratio is the fraction of:*
- Preference Share Capital and Debentures to Equity Share Capital and Reserve & Surplus.*
 - Equity Share Capital and Reserve & Surplus to Preference Share Capital and Debentures.*
 - Equity Share Capital to Total Assets.*
 - Total Assets to Equity Share Capital.*
9. *From the following information, calculate P/E ratio:*
- | | |
|--|--|
| <i>Equity share capital of ` 10 each</i> | <i>`</i> |
| <i>8,00,000</i> | <i>9% Preference share capital of ` 10</i> |
| <i>each</i> | <i>` 3,00,000</i> |
| <i>Profit (after 35% tax)</i> | <i>`</i> |
| <i>2,67,000</i> | |
| <i>Depreciation</i> | <i>` 67,000</i> |
| <i>Market price of equity share</i> | <i>` 48</i> |
- 15 times*
 - 16 times*
 - 17 times*
 - 18 times*
10. *Equity multiplier allows the investor to see:*
- What portion of interest on debt can be covered from earnings available to equity shareholders?*
 - How many times preference share interest be paid from earnings available to equity shareholders?*
 - What portion of return on equity is the result of debt?*
 - How many times equity is multiplied to get the value of debt?*

11. A company has average accounts receivable of ` 10,00,000 and annual credit sales of ` 60,00,000. Its average collection period would be:
- 60.83 days
 - 6.00 days
 - 1.67 days
 - 0.67 days
12. A company has net profit margin of 5%, total assets of ` 90,00,000 and return on assets of 9%. Its total asset turnover ratio would be:
- 1.6
 - 1.7
 - 1.8
 - 1.9
13. What does Q ratio measures?
- Relationship between market value and book value per equity share.
 - Proportion of profit available per equity share.
 - Overall earnings on average total assets.
 - Market value of equity as well as debt in comparison to all assets at their replacement cost.
14. Calculate operating expenses from the information given below:

Sales	` 75,00,000
Rate of income tax	50%
Net profit to sales	5%
Cost of goods sold	` 32,90,000
Interest on debentures	` 60,000

- ` 41,00,000
- ` 8,10,000
- ` 34,00,000
- ` 33,90,000

15. Which of the following is not a profitability ratio?

- (a) P/E ratio
- (b) Return on capital employed (ROCE)
- (c) Q Ratio
- (d) Preference Dividend Coverage Ratio

Theoretical Questions

1. DISCUSS any three ratios computed for investment analysis.
2. DISCUSS the financial ratios for evaluating company performance on operating efficiency and liquidity position aspects.
3. DISCUSS Stock Turnover ratio and Gearing ratio.
4. DISCUSS the composition of Return on Equity (ROE) using the DuPont model.
5. EXPLAIN briefly the limitations of Financial ratios.
6. DISCUSS DuPont Model.

Practical Problems

1. The total sales (all credit) of a firm are ` 6,40,000. It has a gross profit margin of 15 per cent and a current ratio of 2.5. The firm's current liabilities are ` 96,000; inventories ` 48,000 and cash ` 16,000.
 - (a) DETERMINE the average inventory to be carried by the firm, if an inventory turnover of 5 times is expected? (Assume 360 days a year).
 - (b) DETERMINE the average collection period if the opening balance of debtors is intended to be of ` 80,000? (Assume 360 days a year).
2. The capital structure of Beta Limited is as follows:

Equity share capital of ` 10 each	8,00,000
9% preference share capital of ` 10 each	3,00,000
	11,00,000

Additional information: Profit (after tax at 35 per cent) ` 2,70,000; Depreciation ` 60,000; Equity dividend paid 20 per cent; Market price of equity shares ` 40.

You are required to COMPUTE the following, showing the necessary workings:

- (a) Dividend yield on the equity shares
 - (b) Cover for the preference and equity dividends
 - (c) Earnings per shares
 - (d) Price-earnings ratio
3. The following accounting information and financial ratios of PQR Ltd. relates to the year ended 31st March, 2023:

I	Accounting Information:	
	Gross Profit Net profit Raw materials consumed Direct wages Stock of raw materials Stock of finished goods Debt collection period (All sales are on credit)	15% of Sales 8% of sales 20% of works cost 10% of works cost 3 months' usage 6% of works cost 60 days
II	Financial Ratios:	
	Fixed assets to sales Fixed assets to Current assets Current ratio Long-term loans to Current liabilities Share Capital to Reserves and Surplus	1 : 3 13 : 11 2 : 1 2 : 1 1 : 4

If value of Fixed Assets as on 31st March, 2022 amounted to ` 26 lakhs, PREPARE a summarised Profit and Loss Account of the company for the year ended 31st March, 2023 and also the Balance Sheet as on 31st March, 2023.

4. Ganpati Limited has furnished the following ratios and information relating to the year ended 31st March, 2023:

<i>Sales</i>	₹ 60,00,000
<i>Return on net worth</i>	25%
<i>Rate of income tax</i>	50%
<i>Share capital to reserves</i>	7:3
<i>Current ratio</i>	2
<i>Net profit to sales</i>	6.25%
<i>Inventory turnover (based on cost of goods sold)</i>	12
<i>Cost of goods sold</i>	₹ 18,00,000
<i>Interest on debentures</i>	₹ 60,000
<i>Receivables</i>	₹ 2,00,000
<i>Payables</i>	₹ 2,00,000

You are required to:

- CALCULATE the operating expenses for the year ended 31st March, 2023.
- PREPARE a Balance Sheet as on 31st March, 2023 in the following format:

Balance Sheet as on 31st March, 2023

Liabilities	₹	Assets	₹
<i>Share Capital</i>		<i>Fixed Assets</i>	
<i>Reserve and Surplus</i>		<i>Current Assets</i>	
<i>15% Debentures</i>		<i>Stock</i>	
<i>Payables</i>		<i>Receivables</i>	
		<i>Cash</i>	

5. Using the following information, PREPARE the balance sheet:

<i>Long-term debt to net worth</i>	0.5
<i>Total asset turnover</i>	2.5
<i>Average collection period* ₹</i>	18 days

<i>Inventory turnover</i>	9
<i>Gross profit margin</i>	10%
<i>Acid-test ratio</i>	1

Assume a 360-day year and all sales on credit.

	、		、
<i>Cash</i>	?	<i>Notes and payables</i>	1,00,000
<i>Accounts receivable</i>	?	<i>Long-term debt</i>	?
<i>Inventory</i>	?	<i>Common stock</i>	1,00,000
<i>Plant and equipment</i>	?	<i>Retained earnings</i>	1,00,000
<i>Total assets</i>	?	<i>Total liabilities and equity</i>	?

6. *Following information has been provided from the books of Laxmi Pvt. Ltd. for the year ending on 31st March, 2023:*

<i>Net Working Capital</i>	₹ 4,80,000
<i>Bank overdraft</i>	₹ 80,000
<i>Fixed Assets to Proprietary ratio</i>	0.75
<i>Reserves and Surplus</i>	₹ 3,20,000
<i>Current ratio</i>	2.5
<i>Liquid ratio (Quick Ratio)</i>	1.5

You are required to PREPARE a summarised Balance Sheet as at 31st March, 2023 assuming that there is no long term debt.

7. *Manan Pvt. Ltd. gives you the following information relating to the year ending 31st March, 2023:*

(1) <i>Current Ratio 2.5 :</i>	1
(2) <i>Debt-Equity Ratio</i>	1 : 1.5
(3) <i>Return on Total Assets (After Tax)</i>	15%
(4) <i>Total Assets Turnover Ratio</i>	2

(5) Gross Profit Ratio	20%
(6) Stock Turnover Ratio	7
(7) Net Working Capital	` 13,50,000
(8) Fixed Assets	` 30,00,000
(9) 1,80,000 Equity Shares of	` 10 each
(10) 60,000, 9% Preference Shares of	` 10 each
(11) Opening Stock	` 11,40,000

You are required to CALCULATE:

- Quick Ratio
- Fixed Assets Turnover Ratio
- Proprietary Ratio
- Earnings per Share

8. Gig Ltd. has furnished the following information relating to the year ended 31st March, 2022 and 31st March, 2023:

	31st March, 2022 (`)	31<sup>st</sup> March, 2023 (`)
Share Capital	40,00,000	40,00,000
Reserve and Surplus	20,00,000	25,00,000
Long term loan	30,00,000	30,00,000

- Net profit ratio: 8%
- Gross profit ratio: 20%
- Long-term loan has been used to finance 40% of the fixed assets.
- Stock turnover with respect to cost of goods sold is 4.
- Debtors represent 90 days sales.
- The company holds cash equivalent to 1½ months cost of goods sold.
- Ignore taxation and assume 360 days in a year.

You are required to PREPARE Balance Sheet as on 31st March, 2023 in the following format:

Liabilities	(`)	Assets	(`)
Share Capital	-	Fixed Assets	-
Reserve and Surplus	-	Sundry Debtors	-
Long-term loan	-	Closing Stock	-
Sundry Creditors	-	Cash in hand	-

9. Following information relates to Temer Ltd.:

Debtors Velocity	3 months
Creditors Velocity	2 months
Stock Turnover Ratio	1.5
Gross Profit Ratio	25%
Bills Receivables	` 25,000
Bills Payables	` 10,000
Gross Profit	` 4,00,000
Fixed Assets turnover Ratio	4

Closing stock of the period is ` 10,000 above the opening stock. DETERMINE:

- (i) Sales and cost of goods sold
- (ii) Sundry Debtors
- (iii) Sundry Creditors
- (iv) Closing Stock
- (v) Fixed Assets

10. From the following information and ratios, PREPARE the Balance sheet as at 31st March, 2023 and Income Statement for the year ended on that date for M/s Ganguly & Co -

Average Stock	`10 lakh
Current Ratio	3:1
Acid Test Ratio	1:1
PBIT to PBT	2.2:1

Average Collection period (Assume 360 days in a year)	30 days
Stock Turnover Ratio (Use sales as turnover)	5 times
Fixed assets turnover ratio	0.8 times
Working Capital	`10 lakh
Net profit Ratio	10%
Gross profit Ratio	40%
Operating expenses (excluding interest)	` 9 lakh
Long term loan interest	12%
Tax	Nil

11. From the following information, you are required to PREPARE a summarised Balance Sheet for Rudra Ltd. for the year ended 31st March, 2023:

Debt Equity Ratio	1:1
Current Ratio	3:1
Fixed Asset Turnover (on the basis of sales)	4
Stock Turnover (on the basis of sales)	6
Cash in hand	` 5,00,000
Stock to Debtor	1:1
Sales to Net Worth	4
Capital to Reserve	1:2
Gross Profit	20% of Cost
COGS to Creditor	10:1

Interest for entire year is yet to be paid on Long Term loan @ 10%.

Case Scenarios

1. RNOC Ltd is a listed company and has been facing a cash crunch situation since a while. The CFO is of the opinion that excess stock maintained as per the instructions of management of the company

is the reason for cash crunch.

However, the management states that its product line requires larger amount of inventory due to greater variety of product line and customer may ask for any type of product. To maintain competitive advantage, the company should be able to cater to customer needs as and when required. The management is highly critical of the collection team as the management feels that they are not collecting the receivables within time as per industry standards.

You have been hired by the company as a financial consultant. Management has provided you the latest audited financial statements and also relevant industry statistics. You are required to advice the company to improve its liquidity position.

Statement of Profit and Loss		
Sales		1,25,00,000
Cost of goods sold		
Opening Stock	23,00,000	
Add: Purchases	80,00,000	
Add: Direct expenses	12,00,000	
Less: Closing Stock	(38,60,000)	(76,40,000)
Gross Profit		48,60,000
Less: Operating Expenses		
Administrative Expenses	13,20,000	
Selling and Distribution Expenses	15,90,000	(29,10,000)
Operating Profit		19,50,000
Add: Non-Operating Income		3,28,000
Less: Non-Operating Expenses		(1,27,000)
Profit before Interest and taxes		21,51,000
Less: Interest		(4,39,000)
Profit before tax		17,12,000
Less: Taxes		(4,28,000)
Profit after Tax		12,84,000

Balance Sheet

Sources of Funds		`	`
Owned Funds			
	<i>Equity</i>	30,00,000	
	<i>Share Capital</i>		
	<i>Reserves</i>	18,00,000	48,00,000
	<i>and Surplus</i>		
Borrowed Funds			
	<i>Secured Loan</i>	10,00,000	
	<i>Unsecured Loan</i>	4,30,000	14,30,000
Total Funds Raised			62,30,000
Application of Funds			
Non-Current Assets			
	<i>Building</i>	7,50,000	
	<i>Machinery</i>	2,30,000	
	<i>Furniture</i>	7,60,000	
	<i>Intangible Assets</i>	50,000	17,90,000
Current Assets			
	<i>Inventory</i>	38,60,000	
	<i>Receivables</i>	39,97,000	
	<i>ST investments</i>	3,00,000	
	<i>Cash and Bank</i>	2,30,000	83,87,000
Less:			
Current Liabilities			
	<i>Creditors</i>	25,67,000	
			(39,47,000)
	<i>ST loans</i>	13,80,000	
Total Funds Employed			62,30,000

Industry Standards

Receivables Turnover	= 90 Days
Inventory Turnover	= 100 Days
Payables turnover	= 90 Days
Net Asset Turnover	= 4 Times

The company has set certain standards for the upcoming year financial status. All the ratios are based on closing figures in financial statements.

Equity SC to Reserves	= 1
Net Profit Ratio	= 15%
Gross Profit Ratio	= 50%
loan Term Debt to Equity	= 0.5
Debtor Turnover	= 100 Days
Creditor Turnover (based on COGS)	= 100 Days
Inventory	= 70% of Opening inventory

Cash Balance is assumed to remain same for next year.

- (i) *What is the inventory turnover ratio in days and whether assertion of CFO is correct?*
 - (a) *120 days; Assertion of CFO is correct.*
 - (b) *100 days; Assertion of CFO is incorrect.*
 - (c) *185 days; Assertion of CFO is correct.*
 - (d) *150 days; Assertion of CFO is incorrect.*
- (ii) *What is the receivables turnover and whether assertion of management is correct?*
 - (a) *117 days; Assertion of management is correct.*
 - (b) *100 days; Assertion of management is correct.*
 - (c) *85 days; Assertion of management is correct.*
 - (d) *85 days; Assertion of management is not correct.*

- (iii) What is the expense company needs to incur for earning ` 1 of revenue in the last year?
- (a) 0.844
(b) 0.754
(c) 0.962
(d) 0.824
- (iv) What is the projected net working capital of the company?
- (a) ` 42,87,891
(b) ` 40,27,891
(c) ` 48,27,891
(d) ` 48,28,891
- (v) What is the projected Long-Term Debt of the company for the next year?
- (a) ` 60,00,000
(b) ` 30,00,000
(c) ` 14,30,000
(d) ` 28,60,000

ANSWERS/SOLUTION

Answers to the MCQs

1.	(d)	2.	(a)	3.	(c)	4.	(b)	5.	(d)	6.	(c)
7.	(d)	8.	(a)	9.	(b)	10.	(c)	11.	(a)	12.	(c)
13.	(d)	14.	(c)	15.	(d)						

Answers to the Theoretical Questions

1. Please refer paragraph 3.4.2
2. Please refer paragraph 3.4
3. Please refer paragraph 3.3 & 3.2

4. Please refer paragraph 3.4.2
5. Please refer paragraph 6
6. Please refer paragraph 3.4.2

Answers to the Practical Problems

1. (a) Inventory turnover = $\frac{\text{Cost of goods sold}}{\text{Average inventory}}$

Since gross profit margin is 15 per cent, the cost of goods sold should be 85 per cent of the sales.

Cost of goods sold = $0.85 \times \text{` 6,40,000} = \text{` 5,44,000}.$

Thus, $\bullet \frac{\text{` 5,44,000}}{\text{Average inventory}} = 5$

Average inventory = $\frac{\frac{\text{` 5,44,000}}{5}}{5} = \text{` 1,08,800}$

(b) Average collection period = $\frac{\text{Average Receivables}}{\text{Credit Sales}} \times 360 \text{ days}$

Average Receivables = $\frac{(\text{Opening Receivables} + \text{Closing Receivables})}{2}$

Closing balance of receivables is found as follows:

	₹	₹
Current assets (2.5 of current liabilities)		2,40,000
Less: Inventories	48,000	
Cash	16,000	64,000
✓ Receivables		1,76,000

Average Receivables = $\frac{(\text{` 1,76,000} + \text{` 80,000})}{2} = \text{` 1,28,000}$

So, Average collection period = $\frac{\text{` 1,28,000}}{\text{` 6,40,000}} \times 360 = 72 \text{ days}$

2. (a) Dividend yield on the equity shares

$$= \frac{\text{Dividend per share}}{\text{Market price per share}} \times 100 = \frac{₹ 2 \text{ (i.e. } 0.20 \times \frac{₹ 10}{₹ 40})}{₹ 40} \times 100 = 5\%$$

(b) Dividend coverage ratio

$$(i) \text{ Preference} = \frac{\text{Profit after taxes}}{\frac{\text{Dividend payable to preference shareholders}}{\text{shareholders}}} = \frac{₹ 2,70,000}{₹ 27,000 \text{ (i.e. } 0.09 \times ₹ 3,00,000)} = 10 \text{ times}$$

$$(ii) \text{ Equity} = \frac{\text{Profit after taxes} - \text{Preference share dividend}}{\frac{\text{Dividend payable to equity shareholders at current rate of ₹ 2 per share}}{\text{share}}} = \frac{₹ 2,70,000 - ₹ 27,000}{₹ 1,60,000 \text{ (i.e. } 80,000 \text{ shares} \times ₹ 2)} = 1.52 \text{ times}$$

(c) Earnings per equity share = $\frac{\text{Earnings available to equity shareholders}}{\text{Number of equity shares outstanding}}$

$$= \frac{₹ 2,43,000}{80,000} = ₹ 3.04 \text{ per share}$$

(d) Price-earning (P/E) ratio = $\frac{\text{Profit after taxes} - \text{Preference shares dividend}}{\frac{\text{Dividend payable to equity shareholders at current rate of ₹ 2 per share}}{\text{Market price per share}}}$

$$= \frac{₹ 2,43,000}{\frac{₹ 40}{₹ 3.04}} = 13.2 \text{ times}$$

3. (a) Working Notes:

(i) Calculation of Sales = $\frac{\text{Fixed Assets}}{\frac{1}{3} \text{ Sales}} = \frac{₹ 26,00,000}{\frac{1}{3}} = ₹ 78,00,000$

- (ii) Calculation of Current Assets
- | | | |
|---------------------|---|-----------|
| <u>Fixed Assets</u> | • | <u>13</u> |
| Current Assets | | 11 |

$$\frac{26,00,000}{\text{Current Assets}} = \frac{13}{11} \quad \text{Current Assets} = ₹ 22,00,000$$

(iii) Calculation of Raw Material Consumption and Direct Wages

	₹
Sales	78,00,000
Less: Gross Profit @ 15%	11,70,000
Works Cost	66,30,000

Raw Material Consumption (20% of Works Cost) = ₹

13,26,000 Direct Wages (10% of Works Cost) = ₹
6,63,000

(iv) Calculation of Stock of Raw Materials (= 3 months usage)

$$= 13,26,000 \times \frac{3}{12} = ₹ 3,31,500$$

(v) Calculation of Stock of Finished Goods (= 6% of Works Cost)

$$= 66,30,000 \times \frac{6}{100} = ₹ 3,97,800$$

(vi) Calculation of Current Liabilities

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2$$

$$\frac{22,00,000}{\text{Current Liabilities}} = 2 \quad \text{Current Liabilities} = ₹ 11,00,000$$

(vii) Calculation of Receivables

$$\begin{aligned} \text{Average collection period} &= \frac{\text{Receivables}}{\text{Credit Sales}} \times 365 \\ &= \frac{\text{Receivables}}{78,00,000} \times 365 = 60 \end{aligned}$$

$$\text{Receivables} = ₹ 12,82,191.78 \text{ or } ₹ 12,82,192$$

(viii) Calculation of Long term Loan

$$\frac{\text{Long term Loan}}{\text{Current Liabilities}} = \frac{2}{1}$$

$$\frac{\text{Long term loan}}{11,00,000} = \frac{2}{1} \quad \therefore \text{Long term loan} = ₹ 22,00,000.$$

(ix) Calculation of Cash Balance

	₹
Current assets	22,00,000
Less: Receivables	
12,82,192	
Raw materials stock	3,31,500
Finished goods stock	<u>3,97,800</u>
	20,11,492
Cash balance	1,88,508

(x) Calculation of Net worth

Fixed Assets	26,00,000
Current Assets	22,00,000
Total Assets	48,00,000
Less: Long term Loan	22,00,000
Current Liabilities	<u>11,00,000</u>
Net worth	15,00,000

$$\text{Net worth} = \text{Share capital} + \text{Reserves} = 15,00,000$$

$$\text{Also, } \frac{1}{4} = \frac{\text{Share Capital}}{\text{Reserves and Surplus}}$$

$$\text{So, Share capital} = 15,00,000 \times \frac{1}{5} = ₹ 3,00,000 ;$$

$$\text{Reserves and Surplus} = 15,00,000 \times \frac{4}{5} = ₹ 12,00,000$$

**Profit and Loss Account of PQR
Ltd. for the year ended 31st
March, 2023**

Particulars	₹	Particulars	₹
-------------	---	-------------	---

To	Direct Materials	13,26,000	By Sales	78,00,000
"	Direct Wages	6,63,000		
"	Works (Overhead) (Balancing figure)	46,41,000		
"	Gross Profit c/d	11,70,000		
		78,00,000		78,00,000
"	Selling and Distribution Expenses (Balancing figure)	5,46,000	" Gross Profit b/d	11,70,000
"	Net Profit (8% of Sales)	6,24,000		
		11,70,000		11,70,000

**Balance Sheet of PQR
Ltd. as at 31st March,
2023**

Liabilities	∖	Assets	∖
Share Capital	3,00,000	Fixed Assets	26,00,000
Reserves and Surplus	12,00,000	Current Assets:	
Long term loans	22,00,000	Stock of Raw Material	3,31,500
Current liabilities	11,00,000	Stock of Finished Goods	3,97,800
		Receivables	12,82,192
		Cash	1,88,508
	48,00,000		48,00,000

4. (a) Calculation of Operating Expenses for the year ended 31st March, 2023

		(₹)
Net Profit [@ 6.25% of Sales]		3,75,000
Add: Income Tax (@ 50%)		3,75,000

Profit Before Tax (PBT)		7,50,000
<i>Add:</i> Debenture Interest		60,000

Profit before interest and tax (PBIT)		8,10,000
Sales		60,00,000
Less: Cost of goods sold	18,00,000	
PBIT	8,10,000	26,10,000
Operating Expenses		33,90,000

(b) Balance Sheet as on 31st March, 2023

Liabilities	`	Assets	`
Share Capital	10,50,000	Fixed Assets	17,00,000
Reserve and Surplus	4,50,000	Current Assets:	
15% Debentures	4,00,000	Stock	1,50,000
Payables	2,00,000	Receivables	2,00,000
		Cash	50,000
	21,00,000		21,00,000

Working Notes:

- (i) Share Capital and Reserves and Surplus

The return on net worth is 25%. Therefore, the profit after tax of

` 3,75,000 should be equivalent to 25% of the net worth.

$$\frac{\text{Net worth}}{25} \times 100 = \text{` 3,75,000}$$

100

$$\text{Net worth} = \frac{\text{` 3,75,000} \times 100}{25} = \text{` 15,00,000}$$

The ratio of share capital to reserves is 7:3

$$\text{Share Capital} = \text{` 15,00,000} \times \frac{7}{10}$$

•

10 10,50,000

Reserves and Surplus = 15,00,000 - 4,50,000

₹ 3 •

10

(ii) Debentures

Interest on Debentures @ 15% = ₹ 60,000

$$\text{Debentures} = \frac{\text{Interest}}{\text{Rate}} = \frac{60,000}{15\%} = ₹ 4,00,000$$

(iii) Current Assets

Current Ratio = 2

Payables = ₹

2,00,000

$$\text{Current Assets} = 2 \times \text{Current Liabilities} = 2 \times 2,00,000 = ₹ 4,00,000$$

(iv) Fixed Assets

	₹
Share capital	10,50,000
Reserves and Surplus	4,50,000
Debentures	4,00,000
Payables	2,00,000
	21,00,000
Less: Current Assets	4,00,000
Fixed Assets	17,00,000

(v) Composition of Current

Assets Inventory Turnover

= 12

$\frac{\text{Cost of goods sold}}{\text{Closing stock}} = 12$

Closing stock

$$\text{Closing stock} = \frac{\text{Cost of goods sold}}{12} = ₹ 1,50,000$$

Composition

₹

Stock	1,50,000
-------	----------

Receivables	2,00,000
Cash (balancing figure)	50,000
Total Current Assets	4,00,000

5. Working Notes:

(i) Long term Debt

$$0.5 = \frac{\text{Long-term debt}}{\text{Net worth}} = \frac{\text{Long-term debt}}{\text{`1,00,000} + \text{`1,00,000}}$$

$$\text{Long term debt} = \text{`1,00,000}$$

(ii) Total assets

Total liabilities and Equity = Notes and payables + Long-term debt + Common stock + Retained earnings

$$= \text{`1,00,000} + \text{`1,00,000} + \text{`1,00,000} + \text{`1,00,000} = \text{`4,00,000}$$

$$\text{Total assets} = \text{Total liabilities and Equity} = \text{`4,00,000}$$

(iii) Sales and Cost of Goods sold

$$\text{Total asset turnover} = 2.5 = \frac{\text{Sales}}{\text{Total assets}}$$

$$\text{Sales} = \text{`10,00,000}$$

$$\begin{aligned} \text{Cost of goods sold} &= (100\% - \text{Gross Profit margin}) \times \text{Sales} \\ &= (100\% - 10\%) \times \text{`10,00,000} = \text{`9,00,000.} \end{aligned}$$

(iv) Current Assets

$$\text{Inventory turnover} = 9 = \frac{\text{Cost of goods sold}}{\text{Inventory}} = \frac{\text{`9,00,000}}{\text{Inventory}}$$

$$\text{Inventory} = \text{`1,00,000}$$

$$\text{Average collection period} = 18 = \frac{\text{Receivables} \times 360}{\text{Sale}}$$

$$\text{Accounts receivables} = \text{`50,000}$$

$$\text{Acid-test ratio} = 1 = \frac{\text{Cash} + \text{Accounts Receivable}}{\text{Notes and Payables}} = \frac{\text{Cash} + \text{`50,000}}{\text{`1,00,000}}$$

\\ Cash = ` 50,000

(v) Plant and equipment

= Total Assets - Current Assets

= ` 4,00,000 - (` 1,00,000 + ` 50,000 + ` 50,000) = ` 2,00,000

Balance Sheet

	`		`
Cash	50,000	Notes and payables	1,00,000
Accounts receivable	50,000	Long-term debt	1,00,000
Inventory	1,00,000	Common stock	1,00,000
Plant and equipment	2,00,000	Retained earnings	1,00,000
Total assets	4,00,000	Total liabilities and equity	4,00,000

6. Working notes:

(i) Computation of Current Assets and Current Liabilities

$$\frac{\text{Current assets}}{\text{Current liabilities}} = 2.5$$

Current assets = 2.5 Current liabilities

Now, Working capital = Current assets - Current liabilities

` 4,80,000 = 2.5 Current liability - Current

liability Or, 1.5 Current liability = ` 4,80,000

\\ Current Liabilities = ` 3,20,000

So, Current Assets = ` 3,20,000  $\times$  2.5 = ` 8,00,000

(ii) Computation of Inventories

$$\text{Liquid ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

$$1.5 = \frac{\text{Current assets} - \text{Inventories}}{\text{` 3,20,000}}$$

$$1.5 \times \text{ ` 3, 20,000 } = \text{ ` 8,00,000 Inventories}$$

$$\text{Inventories} = \text{ ` 8,00,000 } - \text{ ` 4, 80,000 } = \text{ ` 3,20,000}$$

(iii) Computation of Proprietary fund; Fixed assets; Capital and Sundry creditors

$$\text{Fixed Asset to Proprietary ratio} = \frac{\text{Fixed assets}}{\text{Proprietary fund}} = 0.75$$

$$\text{Fixed Assets} = 0.75 \text{ Proprietary fund}$$

$$\text{Proprietary fund} = \text{Fixed Assets} + \text{Net Working Capital} - \text{Long Term Debt}$$

$$= 0.75 \text{ Proprietary fund} + \text{ ` 4,80,000 } - 0$$

$$\text{Proprietary fund} = \text{ ` 19,20,000}$$

$$\text{and Fixed Assets} = 0.75 \text{ proprietary fund}$$

$$= 0.75 \times \text{ ` 19,20,000 } = \text{ ` 14,40,000}$$

$$\text{Capital} = \text{Proprietary fund} - \text{Reserves \& Surplus}$$

$$= \text{ ` 19,20,000 } - \text{ ` 3,20,000 } = \text{ ` 16,00,000}$$

$$\text{Sundry Creditors} = \text{Current liabilities} - \text{Bank overdraft}$$

$$= \text{ ` 3,20,000 } - \text{ ` 80,000 } = \text{ ` 2,40,000}$$

Balance Sheet as on 31st March, 2023

Liabilities	`	Assets	`
Capital	16,00,000	Fixed Assets	14,40,000
Reserves & Surplus	3,20,000	Inventories	3,20,000
Bank overdraft	80,000	Other Current Assets	4,80,000
Sundry creditors	2,40,000	(Balancing figure)	
	22,40,000		22,40,000

7. Workings Notes:

(i) Computation of Current Assets & Current Liabilities & Total Assets

$$\text{Net Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

$$\begin{aligned}
 &= 2.5 - 1 = 1.5 \\
 \text{Thus, Current Assets} &= \frac{\text{Net Working Capital} \times 2.5}{1.5} \\
 &= \frac{\text{₹}13,50,000 \times 2.5}{1.5} = \text{₹}22,50,000
 \end{aligned}$$

$$\text{Current Liabilities (CL)} = \text{₹}22,50,000 - \text{₹}13,50,000 = \text{₹}9,00,000$$

$$\text{Total Assets} = \text{Current Assets} + \text{Fixed Assets}$$

$$= \text{₹}22,50,000 + \text{₹}30,00,000 = \text{₹}52,50,000$$

(ii) **Computation of Sales & Cost of Goods Sold**

$$\begin{aligned}
 \text{Sales} &= \text{Total Assets Turnover} \times \text{Total Assets} \\
 &= 2 \times (\text{Fixed Assets} + \text{Current Assets}) \\
 &= 2 \times (\text{₹}30,00,000 + \text{₹}22,50,000) \\
 &= \text{₹}1,05,00,000
 \end{aligned}$$

$$\begin{aligned}
 \text{Cost of Goods Sold} &= (100\% - 20\%) \text{ of Sales} = 80\% \text{ of Sales} \\
 &= 80\% \times \text{₹}1,05,00,000 = \text{₹}84,00,000
 \end{aligned}$$

(iii) **Computation of Stock & Quick Assets**

$$\begin{aligned}\text{Average Stock} &= \frac{\text{Cost of Good Sold}}{\text{Stock Turnover Ratio}} = \frac{\text{₹ 84,00,000}}{7} \\ &= \text{₹ 12,00,000} \\ \text{Closing Stock} &= (\text{Average Stock} \times 2) - \text{Opening Stock} \\ &= (\text{₹ 12,00,000} \times 2) - \text{₹ 11,40,000} \\ &= \text{₹ 12,60,000} \\ \text{Quick Assets} &= \text{Current Assets} - \text{Closing Stock} \\ &= \text{₹ 22,50,000} - \text{₹ 12,60,000} = \text{₹ 9,90,000}\end{aligned}$$

(iv) **Computation of Proprietary Fund**

$$\begin{aligned}\text{Debt-Equity Ratio} &= \frac{\text{Debt}}{\text{Equity}} = \frac{1}{1.5} \\ \text{Or, Equity} &= 1.5 \text{ Debt} \\ \text{Total Assets} &= \text{Equity} + \text{Preference capital} + \text{Debt} + \text{CL} \\ \text{₹ 52,50,000} &= 1.5 \text{ Debt} + \text{₹ 6,00,000} + \text{Debt} + \text{₹ 9,00,000} \\ \text{Thus, Debt} &= \frac{\text{₹ 37,50,000}}{2.5} = \text{₹ 15,00,000} \\ \text{Equity} &= \text{₹ 15,00,000} \times 1.5 = \text{₹ 22,50,000} \\ \text{So, Proprietary Fund} &= \text{Equity} + \text{Preference Capital} \\ &= \text{₹ 22,50,000} + \text{₹ 6,00,000} \\ &= \text{₹ 28,50,000}\end{aligned}$$

(v) **Computation of Profit after tax (PAT)**

$$\begin{aligned}&= \text{Total Assets} \times \text{Return on Total Assets} \\ &= \text{₹ 52,50,000} \times 15\% \\ &= \text{₹ 7,87,500}\end{aligned}$$

(a) Quick Ratio

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}} = \frac{\text{₹ 9,90,000}}{\text{₹ 9,00,000}} = 1.1$$

(b) Fixed Assets Turnover Ratio

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Sales}}{\text{Fixed Assets}} = \frac{\text{₹ 1,05,00,000}}{\text{₹ 30,00,000}} = 3.5$$

(c) Proprietary Ratio

$$\text{Proprietary Ratio} = \frac{\text{Proprietary fund}}{\text{Total Assets}} = \frac{\text{₹ 28,50,000}}{\text{₹ 52,50,000}} = 0.54$$

(d) Earnings per Equity Share (EPS)

$$\begin{aligned} \text{Earnings per Equity Share} &= \frac{\text{PAT - Preference Share Dividend}}{\text{Number of Equity Shares}} \\ &= \frac{\text{₹ 7,87,500} - \text{₹ 54,000 (9\% of ₹ 6,00,000)}}{1,80,000} \\ &= \text{₹ 4.075 per share} \end{aligned}$$

8. (i) Change in Reserve & Surplus = ₹ 25,00,000 - ₹ 20,00,000 = ₹ 5,00,000
So, Net profit = ₹ 5,00,000

$$\text{Net Profit Ratio} = 8\%$$

$$\text{Sales} = \frac{\text{₹ 5,00,000}}{8\%} = \text{₹ 62,50,000}$$

- (ii) Cost of Goods sold

$$\begin{aligned} &= \text{Sales} - \text{Gross profit Margin} \\ &= \text{₹ 62,50,000} - 20\% \text{ of ₹ 62,50,000} \\ &= \text{₹ 50,00,000} \end{aligned}$$

- (iii) Fixed Assets = $\frac{\text{₹ 30,00,000}}{40\%} = \text{₹ 75,00,000}$

- (iv) Stock Cost of Goods Sold

= 50,00,000 = ` 12,50,000

Stock Turnover ratio 4

$$(v) \text{ Debtors} = \frac{62,50,000}{360} \times 90 = \text{` 15,62,500}$$

$$(vi) \text{ Cash Equivalent} = \frac{50,00,000}{12} \times 1.5 = \text{` 6,25,000}$$

Balance Sheet as on 31st March 2023

Liabilities	(`)	Assets	(`)
Share Capital	40,00,000	Fixed Assets	75,00,000
Reserve and Surplus	25,00,000	Sundry Debtors	15,62,500
Long-term loan	30,00,000	Closing Stock	12,50,000
Sundry Creditors (Balancing Figure)	14,37,500	Cash in hand	6,25,000
	1,09,37,500		1,09,37,500

9. (i) Determination of Sales and Cost of goods sold:

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Sales}} \times 100$$

$$\text{Or, } \frac{25}{100} = \frac{\text{` 4,00,000}}{\text{Sales}}$$

$$\text{Or, Sales} = \frac{4,00,000}{25} = \text{` 16,00,000}$$

$$\begin{aligned} \text{Cost of Goods Sold} &= \text{Sales} - \text{Gross Profit} \\ &= \text{` 16,00,000} - \text{` 4,00,000} = \text{` 12,00,000} \end{aligned}$$

(ii) Determination of Sundry Debtors:

Debtors' velocity is 3 months or Debtors' collection period is 3 months,

$$\text{So, Debtors' turnover ratio} = \frac{12 \text{ months}}{3 \text{ months}} = 4$$

$$\text{Debtors' turnover ratio} = \frac{\text{Credit Sales}}{\text{Average Accounts Receivable}}$$

$$= \frac{\text{₹ 16,00,000}}{\text{Bills Receivable} + \text{Sundry Debtors}} =$$

Or, Sundry Debtors + Bills receivable = ₹

4,00,000 Sundry Debtors = ₹ 4,00,000 - ₹

25,000 = ₹ 3,75,000

(iii) Determination of Sundry Creditors:

Creditors' velocity of 2 months or credit payment period is 2 months.

$$\text{So, Creditors' turnover ratio} = \frac{12 \text{ months}}{2 \text{ months}} = 6$$

$$\begin{aligned} \text{Creditors turnover ratio} &= \frac{\text{Credit Purchases*}}{\text{Average Accounts Payables}} \\ &= \frac{\text{₹ 12,10,000}}{\text{Sundry Creditors} + \text{Bills Payables}} = \end{aligned}$$

So, Sundry Creditors + Bills Payable = ₹ 2,01,667

Or, Sundry Creditors + ₹ 10,000 = ₹

2,01,667 Or, Sundry Creditors = ₹ 2,01,667 - ₹

10,000 = ₹ 1,91,667

(iv) Determination of Closing Stock

$$\text{Stock Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{\text{₹ 12,00,000}}{\text{Average Stock}} = 1.5$$

So, Average Stock = ₹ 8,00,000

$$\text{Now Average Stock} = \frac{\text{Opening Stock} + \text{Closing Stock}}{2}$$

$$\begin{aligned} \text{Or } \frac{\text{Opening Stock} + (\text{Opening Stock} + \text{₹ 10,000})}{2} &= ₹ 8,00,000 \\ &= \end{aligned}$$

Or, Opening Stock = ₹ 7,95,000

So, Closing Stock = ₹ 7,95,000 + ₹ 10,000 = ₹ 8,05,000

(v) Determination of Fixed Assets

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Fixed Assets}} = 4$$

Fixed Assets

$$\text{Or,} \quad = \frac{\text{`12,00,000}}{\text{Fixed Assets}} =$$

$$\text{Or,} \quad \text{Fixed Asset} = \text{` 3,00,000}$$

Workings:

***Calculation of Credit purchases:**

Cost of goods sold = Opening stock + Purchases - Closing stock

$$\text{` 12,00,000} = \text{` 7,95,000} + \text{Purchases} - \text{` 8,05,000}$$

$$\text{` 12,00,000} + \text{` 10,000} = \text{Purchases}$$

$$\text{` 12,10,000} = \text{Purchases (credit)}$$

Assumption:

- (i) All sales are credit sales
- (ii) All purchases are credit purchase
- (iii) Stock Turnover Ratio and Fixed Asset Turnover Ratio may be calculated either on Sales or on Cost of Goods Sold.

10. 1. Current Ratio = 3:1

Current Assets (CA)/Current Liability (CL)

$$= 3:1 \text{ CA} = 3\text{CL}$$

$$\text{WC} = 10,00,000$$

$$\text{CA} - \text{CL} = 10,00,000$$

$$3\text{CL} - \text{CL} =$$

$$10,00,000$$

$$2\text{CL} = 10,00,000$$

$$\text{CL} = \frac{10,00,000}{2}$$

$$\text{CL} = \text{`5,00,000}$$

$$\text{CA} = 3 \times$$

$$5,00,000 \text{ CA} =$$

$$\text{`15,00,000}$$

$$2. \text{ Acid Test Ratio} = \text{CA} - \text{Stock} / \text{CL} = 1:1$$

$$= \frac{15,00,000 - \text{Stock}}{5,00,000} = 1$$

$$15,00,000 - \text{stock} = 5,00,000$$

$$\text{Stock} = \text{`10,00,000}$$

$$3. \text{ Stock Turnover ratio (on sales)} = 5$$

$$\frac{\text{Sales}}{\text{Avg stock}} = 5$$

$$\frac{\text{Sales}}{10,00,000} = 5$$

$$\text{Sales} = \text{`50,00,000}$$

$$4. \text{ Gross Profit} = 50,00,000 \times 40\% =$$

$$\text{`20,00,000} \quad \text{Net profit (PBT)} = 50,00,000$$

$$\times 10\% = \text{`5,00,000}$$

$$5. \text{ PBIT/PBT} = 2.2$$

$$\text{PBIT} = 2.2 \times 5,00,000$$

$$\text{PBIT} = 11,00,000$$

$$\text{Interest} = 11,00,000 - 5,00,000 = \text{`6,00,000}$$

$$\text{Long term loan} = \frac{6,00,000}{0.12} = \text{`50,00,000}$$

$$0.12$$

$$6. \text{ Average collection period} = 30 \text{ days}$$

$$\text{Receivables} = \frac{30}{360} \times 50,00,000 = 4,16,667$$

$$7. \text{ Fixed Assets Turnover Ratio} =$$

$$0.8 \text{ `50,00,000} / \text{Fixed Assets}$$

$$= 0.8 \text{ Fixed Assets} =$$

$$\text{`62,50,000}$$

Income Statement

	(`)
Sales	50,00,000
Less: Cost of Goods Sold	30,00,000
Gross Profit	20,00,000
Less: Operating Expenses	9,00,000
Less: Interest.	6,00,000
Net Profit	5,00,000

Balance sheet

Liabilities	(`)	Assets	(`)
Equity	22,50,000	Fixed asset	62,50,000
share capital			
Long term debt	50,00,000	Current assets:	
Current liability	5,00,000	Stock 10,00,000	
		Receivables 4,16,667	
		Other <u>83,333</u>	15,00,000
	77,50,000		77,50,000

11.

Balance Sheet of Rudra Ltd.

Liabilities	(`)	Assets	(`)
Capital	10,00,000	Fixed Assets	30,00,000
Reserves	20,00,000	Current Assets:	
Long Term Loan @ 10%	30,00,000	Stock in Trade	20,00,000
Current Liabilities:		Debtors	20,00,000
Creditors	10,00,000	Cash	5,00,000
Other Short-term	2,00,000		

Current Liability (Other STCL)			
Outstanding Interest	3,00,000		
	75,00,000		75,00,000

Working Notes:

Let sales be ` x

Balance Sheet of Rudra Ltd.

Liabilities	(`)	Assets	(`)
Capital		Fixed Assets	x/4
Reserves		Current Assets:	
Net Worth	x/4	Stock in Trade	x/6
Long Term Loan @ 10%	x/4	Debtors	x/6
		Cash	5,00,000
Current liabilities:			
Creditors	x/12		
Other Short-term Current Liability			
Outstanding Interest			
Total Current Liabilities	x/9+5,00,000/3		
Total		Total	

$$1. \text{ Fixed Asset Turnover} = 4 = \frac{x}{\text{Fixed Assets}}$$

$$\text{Fixed Assets} = \frac{x}{4}$$

$$2. \text{ Stock Turnover} = 6 = \frac{x}{\text{stock}}$$

$$\text{Stock} = \frac{x}{6}$$

$$3. \quad \text{Sales to net worth} = 4 = \frac{x}{\text{net worth}}$$

$$\text{Net worth} = \frac{x}{4}$$

$$4. \quad \text{Debt: Equity} = 1 : 1$$

$$\frac{\text{Long Term Loan}}{\text{Net worth}} = \frac{1}{1}$$

$$\text{Long term loan} = \text{Net worth} = \frac{x}{4}$$

$$5. \quad \text{Gross Profit to Cost} = 20\%$$

$$\frac{\text{GP}}{\text{Sales} - \text{G P}} = 20\%$$

$$\frac{\text{G P}}{x - \text{G P}} = 20\%$$

$$\text{GP} = 0.2x - 0.2 \text{ GP}$$

$$1.2 \text{ GP} = 0.2x$$

$$\text{G P} = \frac{0.2x}{1.2}$$

$$\text{G P} = x/6$$

$$\text{Cost of Goods Sold} = x - x/6 = 5/6 x$$

$$6. \quad \text{COGS to creditors} = 10:1$$

$$\frac{\text{COGS}}{\text{Creditors}} = \frac{10}{1}$$

$$\frac{\frac{5}{6}x}{\text{creditors}} = \frac{10}{1}$$

$$\frac{\text{Creditors}}{5x} = \frac{x}{\frac{1}{2}}$$

$$7. \quad \frac{\text{Stock}}{\text{Debtor}} =$$

$$\text{Debtor} = \text{Stock} = \frac{x}{6}$$

$$8. \quad \text{Current Ratio} = 3 : 1$$

$$\frac{\text{Stock} + \text{Debtors}}{\text{Cash Current Liabilities}} = \frac{3}{1}$$

$$\frac{\frac{x}{6} + \frac{x}{6} + 5,00,000}{\text{Current Liabilities}} = 3$$

$$\frac{\frac{x}{3} + 5,00,000}{3} = \text{CL}$$

$$\text{CL} = \frac{x}{9} + \frac{5,00,000}{3}$$

$$9. \quad \text{CA} = 3\text{CL}$$

$$= 3 \left(\frac{x}{9} + \frac{5,00,000}{3} \right)$$

$$\text{CA} = \frac{x}{3} + 5,00,000$$

$$10. \quad \text{Net worth} + \text{Long Term Loan} + \text{Current Liability} = \text{Fixed Asset} + \text{Current Assets}$$

$$\frac{x}{4} + \frac{x}{4} + \frac{x}{9} + \frac{5,00,000}{3} = \frac{x}{4} + \frac{x}{3} + 5,00,000$$

$$\frac{x}{4} + \frac{x}{9} - \frac{x}{3} = 5,00,000 - \frac{5,00,000}{3}$$

$$\frac{9x + 4x - 12x}{36} = \frac{15,00,000 - 5,00,000}{3}$$

$$\frac{x}{36} = \frac{10,00,000}{3}$$

$$x = 1,20,00,000$$

11. Now, from above calculations, we get,

$$\begin{array}{rcl} \text{Fixed Asset} & \frac{\text{₹} 1,20,00,000}{4} & = \text{₹} 30,00,000 \\ = & & \end{array}$$

$$\begin{array}{rcl} \text{Stock} & \frac{\text{₹} 1,20,00,000}{6} & = \text{₹} 20,00,000 \\ = & & \end{array}$$

$$\begin{array}{rcl} \text{Debtor} & \frac{\text{₹} 1,20,00,000}{6} & = \text{₹} 20,00,000 \\ = & & \end{array}$$

$$\begin{array}{rcl} \text{Net Worth} & = \text{₹} 30,00,000 & \\ \text{Now, Capital to Reserve is 1 : 2} & & \\ \text{Capital} & = \text{₹} 10,00,000 & \\ \text{and, Reserve} & = \text{₹} 20,00,000 & \end{array}$$

$$\begin{array}{rcl} \text{Long Term Loan} & = & 30,00,000 \\ = & \times & 4 \end{array}$$

$$\text{Outstanding Interest} = 30,00,000 \times 10\% = 3,00,000$$

$$\begin{array}{rcl} \text{Creditors} & = \frac{\text{₹} 1,20,00,000}{12} & = \text{₹} 10,00,000 \\ = & & \end{array}$$

$$\text{Current Liabilities} = \text{Creditors} + \text{Other STCL} + \text{Outstanding Interest}$$

$$\begin{array}{rcl} \frac{\text{₹} 5,00,000}{9} + \frac{\text{₹} 3,00,000}{3} & = & \text{₹} 10,00,000 + \text{Other STCL} + \text{₹} 3,00,000 \end{array}$$

$$\begin{array}{rcl} \frac{\text{₹} 1,20,00,000}{9} + \frac{\text{₹} 5,00,000}{3} & = & \text{₹} 13,00,000 + \text{Other STCL} \end{array}$$

$$\begin{array}{rcl} \text{₹} 15,00,000 & = & \text{Other STCL} + \text{₹} 13,00,000 \\ \text{Other STCL} & = & \text{₹} 2,00,000 \end{array}$$

Answers to the Case Scenarios

1.

i.	(c)	ii.	(a)	iii.	(a)	iv.	(b)	v.	(b)
-----------	-----	------------	-----	-------------	-----	------------	-----	-----------	-----

i. (c) Inventory Turnover = $\frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{38,60,000}{76,40,000} \times 365$
 = 184.41 days = 185 days
 (apx)

Inventory holding period of 185 days is significantly higher as compared to industry standard of 100 days. This means a significant amount of working capital is tied in inventory, which may be leading to liquidity crunch.

$$\text{ii. (a) Receivables Turnover} = \frac{\text{Receivables}}{\text{Sales}} \times 365 = \frac{39,79,000}{1,25,00,000} \times 365$$

$$= 116.71 = 117 \text{ days (apx)}$$

Receivables turnover of 117 days as compared to industry standard of 90 days is a further delay of 27 days. This will lead to good amount of money being tied up in debtors.

iii. (a) Operating Ratio is the number which indicates cost incurred by company for earning each rupee of revenue

$$\text{Operating Ratio} = \frac{\text{COGS} + \text{Operating Expenses}}{\text{Sales}} \times 100$$

$$= \frac{76,40,000 + 29,10,000}{1,25,00,000} \times 100 = 0.844$$

iv. (b) Equity to Reserves = 1

$$\text{Reserves} = 1 \times 30,00,000 = ₹ 30,00,000$$

$$\text{Projected profit} = 30,00,000 - 18,00,000 = ₹ 12,00,000$$

$$\text{Net Profit Margin} = 15\%$$

$$12,00,000 / \text{Sales} = 0.15$$

$$\text{Sales} = ₹ 80,00,000$$

$$\text{Gross Profit} = 80,00,000 \times 50\% = ₹ 40,00,000$$

$$\text{COGS} = 80,00,000 - 40,00,000 = ₹ 40,00,000$$

Projected Debtors Turnover = 100 days = closing Receivables/

$$100 = \frac{\text{Sales} \times 365}{\text{Closing Receivables}}$$

$$80,00,000 \times 365$$

Closing Receivables = $80,00,000 \times 100 / 365 = \text{` } 21,91,781$
 Projected Closing Inventory = 70% of opening inventory

= $70\% \text{ of } 38,60,000 = \text{` } 27,02,000$

Projected Creditor Turnover = 100 days

= $\text{closing creditors} / \text{COGS} \times 365$

Closing Creditors = $\text{COGS} \times 100 / 365$

Closing Creditor = $40,00,000 \times 100 / 365 = \text{` } 10,95,890$

Net Working Capital =

Cash + Debtors + Inventory -

Creditors

= $2,30,000 + 21,91,781 + 27,02,000 - 10,95,890$

Net Working Capital = $\text{` } 40,27,891$

v. (b) Equity Share Capital + Reserves = $30,00,000 + 30,00,000 = \text{` } 60,00,000$

Long Term Debt to Equity = 0.5

LTD / $60,00,000 = 0.5$

Long Term Debt = $0.5 \times 60,00,000$

Long Term Debt = $\text{` } 30,00,000$

